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COMPACT WAVEGUIDE ILLUMINATION SYSTEM

Abstract

The present application is directed to optical dispatching circuits that may reduce the footprint of an illumination system. In particular, embodiments of the present application provide an illumination system that splits and spreads incoming light sources (e.g., RGB laser light sources) into a plurality of emitters that cover a two-dimensional (2D) area. The present application describes various implementations of an optical dispatching circuit, which receives light as input and is configured to spread this light across a number of waveguides that each emit light from a plurality of locations. The optical dispatching circuits described herein may be configured to receive light from multiple sources emitting at different wavelengths (such as, but not limited to, red, green and blue light) and effectively deliver the light from the multiple sources in a substantially uniform manner to a plurality of emitters that cover a 2D area.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application No. 63/560,408, filed Mar. 1, 2024, titled “Compact Waveguide Splitter Circuit,” the disclosure of which is hereby incorporated, in its entirety, by this reference.

BACKGROUND

[0002] Laser-source-based displays often exhibit better color gamut and higher brightness compared to LEDs. Their narrowband nature also enables a wide range of applications with novel optical designs using diffractive/holographic optical elements or metasurfaces.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The accompanying drawings illustrate a number of exemplary embodiments and are a part of the specification. Together with the following description, these drawings demonstrate and explain various principles of the present disclosure.

[0004] FIG. 1 is a diagram of an illumination system according to some embodiments of this disclosure.

[0005] FIG. 2 depicts an illumination system comprising an evanescent waveguide splitter according to some embodiments of this disclosure.

[0006] FIG. 3 depicts a mode combiner waveguide structure according to some embodiments of this disclosure.

[0007] FIG. 4 is a diagram of a dual injection evanescent waveguide splitter according to some embodiments of this disclosure.

[0008] FIG. 5 depicts an illumination system comprising a nested Y-splitter tree according to some embodiments of this disclosure.

[0009] FIG. 6 is a diagram of an illustrative adiabatic directional coupler according to some embodiments of this disclosure.

[0010] FIG. 7 is a diagram of an illustrative waveguide crossing according to some embodiments of this disclosure.

[0011] FIG. 8 depicts an illumination system comprising a star coupler according to some embodiments of this disclosure.

[0012] FIG. 9 is an illustration of an example artificial-reality system according to some embodiments of this disclosure.

[0013] FIG. 10 is an illustration of an example artificial-reality system with a handheld device according to some embodiments of this disclosure.

[0014] FIG. 11A is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0015] FIG. 11B is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0016] FIG. 12A is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0017] FIG. **12B** is an illustration of example user interactions within an artificial-reality system according to some embodiments of this disclosure.

[0018] FIG. **13** is an illustration of an example wrist-wearable device of an artificial-reality system according to some embodiments of this disclosure.

[0019] FIG. **14** is an illustration of an example wearable artificial-reality system according to some embodiments of this disclosure.

[0020] FIG. **15** is an illustration of an example augmented-reality system according to some embodiments of this disclosure.

[0021] FIG. **16A** is an illustration of an example virtual-reality system according to some embodiments of this disclosure.

[0022] FIG. **16B** is an illustration of another perspective of the virtual-reality systems shown in FIG. **16A**.

[0023] FIG. **17** is a block diagram showing system components of example artificial- and virtual-reality systems.

[0024] FIG. **18** is an illustration of an example system that incorporates an eye-tracking subsystem capable of tracking a user's eye(s).

[0025] FIG. **19** is a more detailed illustration of various aspects of the eye-tracking subsystem illustrated in FIG. **18**.

[0026] Throughout the drawings, identical reference characters and descriptions indicate similar, but not necessarily identical, elements. While the exemplary embodiments described herein are susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. However, the exemplary embodiments described herein are not intended to be limited to the particular forms disclosed. Rather, the present disclosure covers all modifications, equivalents, and alternatives falling within the scope of the appended claims.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0027] Laser-source-based displays often exhibit better color gamut and higher brightness compared to LEDs. For example, a photonic integrated circuit (PIC)-based front/backlight illuminator may offer advantages such as low power consumption, structural compactness, and high controllability of the emitted light with respect to numerical aperture (NA), chief ray angle, polarization as well as emission cone angle. However, a large overhead area is often required for the part of optical dispatching circuit compared to the active display area, which may significantly increase the fabrication cost of PIC devices.

[0028] The present application is directed to optical dispatching circuits that may reduce the footprint of an illumination system. In particular, embodiments of the present application provide an illumination system that splits and spreads incoming light sources (e.g., RGB laser light sources) into a plurality of emitters that cover a two-dimensional (2D) area. In some embodiments, the light sources may be split and spread across the emitters with a low insertion loss and/or a high uniformity. The present application describes various implementations of an optical dispatching circuit, which receives light as input and is configured to spread this light across a number of waveguides that each emit light from a plurality of locations. The optical dispatching circuits described herein may be configured to receive light from multiple sources emitting at different wavelengths (such as, but not limited to, red, green and blue light) and effectively deliver the light from the multiple sources in a substantially uniform manner to a plurality of emitters that cover a 2D area.

[0029] FIG. **1** is a diagram of an illumination system according to some embodiments of this disclosure. In the example of FIG. **1**, illumination system **100** comprises a display area **112** for emitting light and an optical dispatching circuit **120**, which receives light inputs **110a**, **110b** and **110c**. The illumination system **100** may for instance, be part of a display for a device such as a head-mounted unit, AR/VR or other artificial reality, mixed reality, augmented reality, extended

reality, device, mobile device, etc.

[0030] In the example of FIG. 1, the display area **112** comprises a plurality of waveguides **114** arranged on a substrate (e.g., transparent glass or quartz substrate). The waveguides in the example of FIG. 1 are arranged in groups of three waveguides, one for each of the colors of light input as inputs **110a**, **110b** and **110c**. The light inputs **110a**, **110b** and **110c** may, for instance, provide red, green and blue light, respectively, and the optical dispatching circuit is configured to direct light from these inputs to one of the waveguides in each group of three waveguides. As such, in this example for a given group of three waveguides, red light would propagate through one of the waveguides, green light would propagate through another one of the waveguides, and blue light would propagate through the other waveguide. As illustrated in FIG. 1, the waveguides **114** may be arranged in a particular sequence (e.g., RGB) which may repeat such that rows **116** may be established across the waveguide columns **114**. Any number of waveguide rows and columns may be provided in the illumination system **100**, and any number of emitters may be provided in each waveguide.

[0031] In the example of FIG. 1, each of the waveguides **114** comprises a plurality of emitters as shown in insert **118** which represents the region **117**. The emitters in this example are grating emitters, and may be provided along the length of each of the waveguides. As such, the emitters may be arranged over the display area **112** and configured to emit light across this area when light propagates through the respect waveguide. A cross-sectional view through three of the waveguides **114** is shown in inset **119**, illustrating that the waveguides are arranged as structures on top of a substrate **121**.

[0032] In summary, the light input to the optical dispatching circuit **120** is distributed, by the optical dispatching circuit, to the waveguides **114**, with the input light **110a** being distributed to a first set of waveguides, the input light **110b** being distributed to a second set of waveguides and the input light **110c** being distributed to a third set of waveguides of the display area **112**. Emitter structures are formed in each of the waveguides **114** that produce light from the waveguides perpendicular to the light propagation direction in the waveguide. In the example of FIG. 1, the light propagates along the y-axis in the waveguides, and is emitted in a direction coming out of the page (z-axis).

[0033] In some embodiments, the light inputs **110a**, **110b** and **110c** each comprise a laser light source. Additionally, or alternatively, the light inputs **110a**, **110b** and **110c** may comprise a respective waveguide that directs light into the optical dispatching circuit **120**, and which may receive light from a light source separate from the illumination system **100** (e.g., laser sources are connected to waveguides of the illumination system **100**, which provide light inputs to the optical addressing circuit).

[0034] In some embodiments, a width of each of the waveguides **114** (e.g., in the example of FIG. 1 the size of the waveguide in the x-direction) is greater than or equal to 1 μm , 1.5 μm , 2 μm , 2.5 μm or 3 μm . In some embodiments, the width of each of the waveguides **114** is less than or equal to 3.5 μm , 3 μm , 2.5 μm , 2 μm or 1.5 μm . Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of each of the waveguides **114** is greater or equal to 1.5 μm and less than or equal to 2.5 μm , etc.).

[0035] To produce a display, the illumination system **100** may be combined with a suitable panel that is configured to allow the light from the illumination system to pass through, or be blocked by, individual pixels. For instance, a display may comprise the illumination system **100** arranged next to a liquid crystal on silicon (LCOS) panel. Light from the emitters of the illumination system **100** may be incident on any number of pixels in a display panel, including a large number of pixels (e.g., hundreds). That is, the emitters of the illumination system **100** may not be pixels, but rather emitters that produce diverging light rays that together cover the surface of a display panel. In such a configuration, the global uniformity of the light produced may be more important than the relative brightness of light produced by the emitters. That is, if some emitters emit more light than

other emitters, this may not present an issue so long as the light combined from all the emitters produces a suitably uniform light field.

[0036] Hereinafter, the three light inputs **110a**, **110b** and **110c** to the optical dispatching circuit **120** will be described as being red, green and blue light for purposes of illustration. However, it will be appreciated that these three light inputs may in general be configured to accept light of any three different wavelengths, and the techniques described herein are not necessarily limited to the use of any particular wavelengths or colors of light.

[0037] Described below are various approaches for implementing the optical dispatching circuit **120**. Each of these approaches may be utilized in the system of FIG. 1 or in any other suitable system in which light from multiple sources are directed to a plurality of waveguides, with light from a given source being directed to one subset of the waveguides.

[0038] FIG. 2 depicts an illumination system comprising an evanescent waveguide splitter according to some embodiments of this disclosure. In the example of FIG. 2, the optical dispatching circuit **120** comprises bus waveguide **211** and transport slab **212**. Light **110**, which may include light from light inputs **110a**, **110b** and **110c** combined together, is input to the bus waveguide **211** and propagates through the bus waveguide. According to some embodiments, the bus waveguide may be implemented as a photonic integrated circuit (PIC). The light in the bus waveguide couples across the gap to the transport slab **212**, which is arranged adjacent to and apart from the bus waveguide, and as a result the bus waveguide injects the different wavelengths of light together into the transport slab. The light propagates through the transport slab and into a plurality of waveguide receivers that include the labeled waveguide receivers **214a** and **214d**, which propagate the light into respective coarse-wavelength multiplexers (CWMs) **220**. In the implementation of optical dispatching circuit **120** shown in FIG. 2, the CWMs **220** split the light into different frequency components, which propagate through three distinct waveguides coupled to a given CWM. Slab **210** is optional and may not be included in all implementations. In some examples, to further reduce footprint, CWMs **220** may be a directional coupler or an inverse designed pattern.

[0039] As shown in FIG. 2, the gap between the bus waveguide **211** and the transport slab **212** narrows with distance (along the y-direction) from the input to the optical dispatching circuit **120**. For instance, the gap between the bus waveguide **211** and the transport slab **212** proximate to receiver **214a** is smaller than the gap between the waveguides proximate to receiver **214d**. This configuration may provide for a relative consistent amount of light propagating into each of the different waveguides receivers. For instance, the amount of light propagating through the bus waveguide **211** will generally decrease with increasing distance along the bus waveguide in the y-direction. As a result, with a constant gap between the bus waveguide **211** and the transport slab **212**, the amount of light that would propagate through receiver **214a** would be less than receiver **214d**. By reducing the gap between the waveguides, the fraction of the light in the bus waveguide that leaks into the transport slab increases. As such, by tuning the size of the gap along the length of the bus waveguide, a uniform transfer of light into the different receivers may be achieved.

[0040] According to some embodiments, different wavelengths of light may leak from the bus waveguide **211** into the transport slab **212** at different rates. As such, tuning the size of the gap along the bus waveguide may not be sufficient to direct a uniform amount of light into each waveguide, because some waveguides may receive more light at certain wavelengths if those wavelengths more readily leak from the bus waveguide into the transport slab than other wavelengths. For instance, red light may be expected to leak from the bus waveguide into the transport slab at a higher rate than blue light. The inventors have recognized that light **110** may be configured with different frequencies of light provided in different modes, which provide for substantially uniform coupling of the different frequencies from the bus waveguide into the transport slab. For example, light **110** may comprise red light in a first mode (e.g., the zeroth fundamental mode), green light in a second mode (e.g., the first fundamental mode), and blue light

in a third mode (e.g., the second fundamental mode), where the first, second and third modes are different modes of light. According to some embodiments, light **110** may comprise red light in a fundamental 0th mode, green light in a 1st higher-order mode, and blue light in a 2nd higher-order mode. The optical dispatching circuit **120** as shown in FIG. 2 may advantageously exhibit significant footprint reduction compared to other optical circuit designs.

[0041] According to some embodiments, optical dispatching circuit **120** shown in FIG. 2 may include an additional CWM coupled to the bus waveguide **211** and configured to input the light **110** into the bus waveguide. The additional CWM (not shown in FIG. 2) may be configured to receive multiple wavelengths of light and to adjust the modes of these wavelengths to produce different frequencies of light provided in different modes, as described above.

[0042] According to some embodiments, an additional CWM, whether included in optical dispatching circuit **120** or provided separately, may create different modes of light that are input into the bus waveguide **211** may comprise an asymmetric directional coupler. An example of such a CWM is shown in FIG. 3.

[0043] In the example of FIG. 3, CWM **300** comprises three waveguides **311**, **312** and **313**. Multiple lasers may be separately coupled to the waveguides **311**, **312** and **313** in the fundamental mode. For instance, the CWM **300** may comprise, or may be coupled to, a red laser, a green laser, and a blue laser which direct red, green and blue light, respectively, in the fundamental mode, into the waveguides **311**, **312** and **313**, respectively.

[0044] The illustrative CWM **300** is configured to combine light from three light sources of different wavelengths, propagating in separate waveguides, into a single waveguide where the light exhibits three different modes. In the example of FIG. 3, green light is propagating in the waveguide **311**, red light in the waveguide **312**, and blue light in the waveguide **313**. In the section **301a**, shown in more detail as inset **301b**, the green light in waveguide **311** is combined with the red light in the waveguide **312** through a directional coupler. The directional coupler may be designed with two close waveguides of different widths, and the effective indices of the waveguides **311** and **312** may be carefully tuned, such that the green light is transferred from the fundamental mode of waveguide **311** to the 1st higher order mode in waveguide **312** (e.g., with near-unity conversion efficiency). The Red light may remain the fundamental mode in waveguide **312**. Similarly, as in the section **302a**, shown in more detail as inset **302b**, blue light in waveguide **313** may be transferred from the fundamental mode of waveguide **313** to the 2nd higher-order mode in waveguide **312**, while the red and green light in the waveguide **312** remain in the fundamental and 1st higher order modes, respectively. As a result, the CWM may produce red, green and blue light together, in different modes. This light may be provided as input light **110** to the bus waveguide **211** as shown in FIG. 2 and described above.

[0045] According to some embodiments, any one or more of the waveguides **311**, **312** and **313** may be tapered in width in one or both of the sections **301a** and **302a** to produce a desired transfer of the light from one of waveguides **311** or **313** into the waveguide **312**. For example, in section **301a**, the waveguide **311** may be tapered to be decreasing in width from left to right as shown in FIG. 3, while the waveguide **312** may optionally be tapered to be increasing in width from left to right. By tuning the change in width and size of the gap between the waveguides **311** and **312**, the fundamental mode of light in the waveguide **311** may couple to the 1st higher order mode of light in the waveguide **312** with a very high (e.g., near unity) transition efficiency. Similarly, in section **302a**, the waveguide **313** may be tapered to be decreasing in width from left to right as shown in FIG. 3, while the waveguide **312** may optionally be tapered to be increasing in width from left to right. By tuning the change in width and size of the gap between the waveguides **313** and **312**, the fundamental mode of light in the waveguide **313** may couple to the 2nd higher order mode of light in the waveguide **312**. It may be noted in the example of FIG. 3 that the waveguides **311**, **312** and **313** do not touch, but rather come close to each other and taper narrower in the sections **301a** and **302a**.

[0046] In some embodiments, a width of waveguide **311** is greater than or equal to 350 nm, 400 nm, 450 nm, 500 nm, 550 nm or 600 nm. In some embodiments, the width of waveguide **311** is less than or equal to 650 nm, 600 nm, 550 nm, 500 nm, 450 nm, or 400 nm. Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of waveguide **311** is greater or equal to 400 nm and less than or equal to 600 nm, etc.). The width of the waveguide **311** may, in some embodiments, narrow along the section **301a** as described above. For instance, the width may narrow from 550 nm to 450 nm.

[0047] In some embodiments, a width of waveguide **313** is greater than or equal to 200 nm, 250 nm, 300 nm, 350 nm, or 400 nm. In some embodiments, the width of waveguide **313** is less than or equal to 450 nm, 400 nm, 350 nm, 300 nm, or 250 nm. Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of waveguide **313** is greater or equal to 250 nm and less than or equal to 400 nm, etc.). The width of the waveguide **313** may, in some embodiments, narrow along the section **301b** as described above. For instance, the width may narrow from 350 nm to 270 nm.

[0048] In some embodiments, a width of waveguide **312** is greater than or equal to 900 nm, 1 μm , 1.1 μm , 1.2 μm , 1.3 μm or 1.4 μm . In some embodiments, the width of waveguide **312** is less than or equal to 1.5 μm , 1.4 μm , 1.3 μm , 1.2 μm , 1.1 μm or 1 μm . Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of waveguide **312** is greater or equal to 1.1 μm and less than or equal to 1.3 μm , etc.). The width of the waveguide **312** may, in some embodiments, widen along the section **301a** or section **301b** as described above. For instance, the width may widen from 1.15 μm to 1.25 μm .

[0049] In the example of FIG. 2, the amount of light emitted from the emitters of the waveguides **114** in display area **112** may decrease along the x-direction due to the amount of light in the waveguides gradually decreasing as more and more is emitted from the waveguide through the emitters. In some embodiments, the optical dispatching circuit **120** may be configured with multiple instances of the waveguides shown in FIG. 2, where the receivers of the optical dispatching circuit **120** are arranged at opposing ends of the waveguides **114** in the display area **112**.

[0050] FIG. 4 is a diagram of a dual injection evanescent waveguide splitter that utilizes this approach, according to some embodiments of this disclosure. In the example of FIG. 4, the optical dispatching circuit **120** comprises two instances of the bus waveguide **211** and transport slab **212**, in addition to two corresponding sets of receivers and CWMs **220**. By injecting light into the waveguides **114** in display area **112** at both ends, the emitters in the display area **112** may more evenly produce light.

[0051] FIG. 5 depicts an illumination system comprising a nested Y-splitter tree according to some embodiments of this disclosure. FIG. 5 depicts another illustrative design for optical dispatching circuit **120**. In the example of FIG. 5, the optical dispatching circuit **120** comprises a plurality of waveguides arranged in a tree structure. As shown, the tree **501** receives light **110a**, **110b** and **110c** into three separate waveguides, which each split into two branches at a Y-split structure. The tree **501** also includes a plurality of waveguide crossers **511**, shown as rectangles in the tree (only one is labeled in FIG. 5).

[0052] In the example of FIG. 5, each waveguide crosser **511** couples to two waveguides as inputs and two waveguides as outputs. In operation, the waveguide crosser receives light of two different wavelengths along the two input waveguides, and outputs the received light from the two output waveguides, with the light inputs spatially swapped. That is, the light provided to the left side input of a waveguide crosser may be output from the right side output of the waveguide crosser, and vice versa. The waveguide crossers allow for the light inputs **110a**, **110b** and **110c** at each wavelength to be output from a plurality of waveguides at the other end of the tree **501** by allowing the paths of a given wavelength of light to effectively pass through or over the paths of a different wavelength of light. For example, the labeled waveguide crosser **511** has light **110b** (e.g., green) and light **110a**

(e.g., red) as inputs on the left and the right, respectively. This waveguide crosser **511** then outputs red light from the left output and green light from the right output, effectively crossing the red and green light paths.

[0053] As shown in the example of FIG. 5, the tree **501** outputs are a plurality of waveguides arranged in a series with a set of each input color being arranged as a group next to each other. For example, as shown in FIG. 5, the output waveguides propagate light in the following order from left to right: blue, green, red, blue, green, red, blue, green, red, blue, green and red. The tree **501** may be arranged with any number of output waveguides by selecting an appropriate number of waveguide Y-split structures and waveguide crossers.

[0054] In the example of FIG. 5, irrespective of the number of waveguides arranged at the output of tree **501**, the light from these waveguides is directed through an array of waveguides **502** into the display area **112**, which produces light from the emitters along the waveguides as shown and as described above. The optical dispatching circuit **120** shown in FIG. 5 may provide for highly uniform light across the rows of waveguides **114** in the display area **112** since the input light may be evenly split at the Y-split structures and crossed without significant loss in the waveguide crossers **511**.

[0055] FIG. 6 is a diagram of an illustrative directional coupler that may be suitable for use as a waveguide crosser **511**, according to some embodiments of this disclosure. In the example of FIG. 6, the directional coupler **600** is configured to swap the spatial modes of light provided at the two inputs to the directional coupler. As shown, directional coupler **600** is left-right symmetrical and includes two waveguides **601** and **602** that taper in width from one side to the other. In particular, waveguide **601** widens from the input to the output side, and waveguide **602** narrows from the input to the output side. The widths of the waveguides at the input side may be different from one another, and the widths of the waveguides at the output side may be different from one another.

[0056] According to some embodiments, the dimensions of the waveguides **601** and **602** may be tuned so that the light input to waveguide **601** (**602**) will be transmitted to waveguide **602** (**601**). While different wavelengths of light may be transmitted in a different manner, the inventors have recognized that with a sufficiently long coupler and thus a sufficiently slow tapered waveguide, the directional coupler will produce high (e.g., close to 100%) power transfer of light that is independent of light frequency. In some embodiments, the length of the tapered section of the waveguides **608** is between 500 μm and 750 μm . The gap between the waveguides may be between 400 nm and 600 nm.

[0057] FIG. 7 is a diagram of an illustrative waveguide crossing that may be suitable for use as waveguide crosser **511**, according to some embodiments of this disclosure. A more compact waveguide crossing may be realized through an inverse design with the technique of shape optimization of the waveguide. In the example of FIG. 7, inverse-designed waveguide crossing **700** is a waveguide shaped as a crossing between two inputs and two outputs. For example, light provided to input **701** through the left may be routed to the output **703**, whereas light provided to input **702** may be routed to the output **704**.

[0058] FIG. 8 depicts an illumination system comprising a star coupler according to some embodiments of this disclosure. FIG. 8 depicts another illustrative design for optical dispatching circuit **120**. In the example of FIG. 8, light **110a**, **110b** and **110c** is input to three separate waveguides **801**, **802** and **803**, respectively, which terminate with tips **804** that are arranged in close proximity to one another. Light from the waveguides **801**, **802** and **803** propagates into the slab waveguide **805** and spreads out radially in the slab waveguide. The light from the slab waveguide is then collected by a large array of receiver waveguides arranged at the opposing side of the tips **804**. The receiver waveguides include illustrative waveguides **806** and **807**, and the receiver waveguides collect light of each wavelength represented by light **110a**, **110b** and **110c**. The receiver waveguides are coupled via an array of waveguides **810** to waveguides **811**, which are coupled to CWMs **820**. The CWMs **820** split the incoming light into the wavelengths of the input light and

direct this light into the waveguides of the display area **112**. The number of receiver waveguides is the same as the number of waveguides **811** and CWMs **820**.

[0059] As shown in FIG. **8**, the receiver waveguides are arranged to be wider the further from the central axis of the slab waveguide **805** they are located. Since the light propagates from the tips **804** into the slab waveguide **805**, more light will propagate in the center of the slab waveguide, and less light will propagate the further outward from the central axis the receiver waveguide is placed (e.g., the received light power may be Gaussian with angle of propagation from the tip to the receiver waveguides). As such, the receiver waveguides may be gradually larger so that each receiver waveguide collects substantially the same amount of light, and consequently the waveguides **114** of display area **112** receive substantially the same amount of light as one another.

[0060] In some embodiments, a width of the waveguides **801**, **802** and **803** at the tip ends **804** is greater than or equal to 0.8 μm , 0.9 μm , 1 μm , 1.1 μm , or 1.2 μm . In some embodiments, the width of the waveguides **801**, **802** and **803** at the tip ends **804** is less than or equal to 1.3 μm , 1.2 μm , 1.1 μm , 1 μm , or 0.9 μm . Any suitable combinations of the above-referenced ranges are also possible (e.g., the width of the waveguides **801**, **802** and **803** at the tip ends **804** is greater or equal to 0.9 μm and less than or equal to 1.1 μm , etc.). In some embodiments, the waveguides **801**, **802** and **803** widen at the tip ends **804** compared with the width of the waveguides leading up to the tip ends **804** (e.g., the waveguides are 0.7 μm wide and widen to 1 μm wide at the tips).

[0061] In some embodiments, the slab waveguide **805** is a 0.5 mm to 2 mm long waveguide shaped as an arc or annular arc (e.g., a half-circle or half-circle annulus).

[0062] Features from any of the embodiments described herein may be used in combination with one another in accordance with the general principles described herein. These and other embodiments, features, and advantages will be more fully understood upon reading the following detailed description in conjunction with the accompanying drawings and claims.

EXAMPLE EMBODIMENTS

[0063] Example 1. A device comprising: a plurality of waveguides consisting of a first plurality of waveguides, a second plurality of waveguides and a third plurality of waveguides; and an optical dispatching circuit comprising a first input, a second input and a third input, the optical dispatching circuit configured to direct light from the first input through each of the first plurality of waveguides, to direct light from the second input through each of the second plurality of waveguides, and to direct light from the third input through each of the third plurality of waveguides.

[0064] Example 2. The device of example 1, wherein the optical dispatching circuit comprises: an input multiplexer coupled to the first input, the second input and the third input and comprising an output; a bus waveguide coupled to the output of the input multiplexer; and a transport slab arranged adjacent to and apart from the bus waveguide, the transport slab coupled to each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.

[0065] Example 3. The device of example 2, wherein a distance between the bus waveguide and the transport slab narrows from a first end of the bus waveguide proximate to the input multiplexer to an opposing end of the bus waveguide distal from the input multiplexer.

[0066] Example 4. The device of example 2, further comprising a plurality of output multiplexers that couple the transport slab to the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides, wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.

[0067] Example 5. The device of example 2, wherein the input multiplexer comprises: a first multiplexer waveguide coupled to the first input; a second multiplexer waveguide coupled to the second input; and a third multiplexer waveguide coupled to the third input and coupled to the output of the input multiplexer, wherein the first multiplexer waveguide widens from the first input

to the output of the input multiplexer, wherein the second multiplexer waveguide narrows from the second input to the output of the input multiplexer, and wherein the third multiplexer waveguide narrows from the third input to the output of the input multiplexer.

[0068] Example 6. The device of example 1, wherein the optical dispatching circuit comprises a waveguide tree coupled to the first input, the second input and the third input, and comprising a plurality of outputs that includes an output for each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.

[0069] Example 7. The device of example 6, wherein the waveguide tree comprises a plurality of Y-shaped waveguide sections.

[0070] Example 8. The device of example 6, wherein the waveguide tree comprises a plurality of directional couplers, each directional coupler configured to swap spatial modes of light across a pair of waveguides therein.

[0071] Example 9. The device of example 8, wherein, in each directional coupler of the plurality of directional couplers, one of the pair of waveguides tapers narrower across the directional coupler from an input side to an output side while another of the pair of waveguides tapers wider from the input side to the output side of the directional coupler.

[0072] Example 10. The device of example 1, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.

[0073] Example 11. The device of example 10, further comprising a plurality of output multiplexers that couple each of the plurality of receiving waveguides to the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides, wherein each receiving waveguide of the plurality of receiving waveguides is coupled to a respective output multiplexer of the plurality of output multiplexers, and wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.

[0074] Example 12. The device of example 10, wherein a width of each receiving waveguide of the plurality of receiving waveguides increases with distance from a central axis of the coupler waveguide.

[0075] Example 13. The device of example 10, further comprising: a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide; a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide; a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide, wherein the first tip, the second tip and the third tip each have a width between 0.9 μm and 1.1 μm .

[0076] Example 14. The device of example 1, further comprising: a first light source arranged to direct light of a first wavelength through the first input of the optical dispatching circuit; a second light source arranged to direct light of a second wavelength, different to the first wavelength, through the second input of the optical dispatching circuit; and a third light source arranged to direct light of a third wavelength, different to the first wavelength and the second wavelength, through the third input of the optical dispatching circuit.

[0077] Example 15. The device of example 14, wherein the first wavelength is between 600 nm and 700 nm, the second wavelength is between 500 nm and 600 nm, and the third wavelength is between 400 nm and 500 nm.

[0078] Example 16. The device of example 1, wherein each of the plurality of waveguides comprises a plurality of emitter structures configured to output light.

[0079] Example 17. A display comprising the device of example 16 and a liquid crystal on silicon (LCoS) panel arranged to receive light output from the plurality of emitter structures.

[0080] Example 18. The display of example 17 wherein the LCOS panel comprises a plurality of

pixels each arranged to receive light from a plurality of the emitter structures.

[0081] Example 19. A method comprising: directing light from a first light source into a first input of an optical dispatching circuit, through the optical dispatching circuit, into a first plurality of waveguides and into a first plurality of emitter structures arranged within the first plurality of waveguides; directing light from a second light source into a second input of the optical dispatching circuit, through the optical dispatching circuit, into a second plurality of waveguides and into a second plurality of emitter structures arranged within the second plurality of waveguides; and directing light from a third light source into a third input of the optical dispatching circuit, through the optical dispatching circuit, into a third plurality of waveguides and into a third plurality of emitter structures arranged within the third plurality of waveguides.

[0082] Example 20. The method of example 19, wherein the first light source produces light having a wavelength between 600 nm and 700 nm, wherein the second light source produces light having a wavelength between 500 nm and 600 nm, and wherein the third light source produces light having a wavelength between 400 nm and 500 nm.

[0083] Example 21. The method of example 19, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.

[0084] Example 22. The method of example 21, wherein: directing light from the first light source into the first input of the optical dispatching circuit comprises directing the light from the first light source through a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide; directing light from the second light source into the second input of the optical dispatching circuit comprises directing the light from the second light source through a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide; and directing light from the third light source into the third input of the optical dispatching circuit comprises directing the light from the third light source through a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide.

[0085] Example 23. The method of example 22, wherein the first tip, the second tip and the third tip each have a width between 0.9 μm and 1.1 μm .

[0086] Embodiments of the present disclosure may include or be implemented in conjunction with various types of Artificial-Reality (AR) systems. AR may be any superimposed functionality and/or sensory-detectable content presented by an artificial-reality system within a user's physical surroundings. In other words, AR is a form of reality that has been adjusted in some manner before presentation to a user. AR can include and/or represent virtual reality (VR), augmented reality, mixed AR (MAR), or some combination and/or variation of these types of realities. Similarly, AR environments may include VR environments (including non-immersive, semi-immersive, and fully immersive VR environments), augmented-reality environments (including marker-based augmented-reality environments, markerless augmented-reality environments, location-based augmented-reality environments, and projection-based augmented-reality environments), hybrid-reality environments, and/or any other type or form of mixed- or alternative-reality environments.

[0087] AR content may include completely computer-generated content or computer-generated content combined with captured (e.g., real-world) content. Such AR content may include video, audio, haptic feedback, or some combination thereof, any of which may be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional (3D) effect to the viewer). Additionally, in some embodiments, AR may also be associated with applications, products, accessories, services, or some combination thereof, that are used to, for example, create content in an artificial reality and/or are otherwise used in (e.g., to perform activities in) an artificial reality.

[0088] AR systems may be implemented in a variety of different form factors and configurations.

Some AR systems may be designed to work without near-eye displays (NEDs). Other AR systems may include a NED that also provides visibility into the real world (such as, e.g., augmented-reality system **1500** in FIG. **15**) or that visually immerses a user in an artificial reality (such as, e.g., virtual-reality system **1600** in FIGS. **16A** and **16B**). While some AR devices may be self-contained systems, other AR devices may communicate and/or coordinate with external devices to provide an AR experience to a user. Examples of such external devices include handheld controllers, mobile devices, desktop computers, devices worn by a user, devices worn by one or more other users, and/or any other suitable external system.

[0089] FIGS. **9-12B** illustrate example artificial-reality (AR) systems in accordance with some embodiments. FIG. **9** shows a first AR system **900** and first example user interactions using a wrist-wearable device **902**, a head-wearable device (e.g., AR glasses **1500**), and/or a handheld intermediary processing device (HIPD) **906**. FIG. **10** shows a second AR system **1000** and second example user interactions using a wrist-wearable device **1002**, AR glasses **1004**, and/or an HIPD **1006**. FIGS. **11A** and **11B** show a third AR system **1100** and third example user **1108** interactions using a wrist-wearable device **1102**, a head-wearable device (e.g., VR headset **1150**), and/or an HIPD **1106**. FIGS. **12A** and **12B** show a fourth AR system **1200** and fourth example user **1208** interactions using a wrist-wearable device **1230**, VR headset **1220**, and/or a haptic device **1260** (e.g., wearable gloves).

[0090] A wrist-wearable device **1300**, which can be used for wrist-wearable device **902**, **1002**, **1102**, **1230**, and one or more of its components, are described below in reference to FIGS. **13** and **14**; head-wearable devices **1500** and **1600**, which can respectively be used for AR glasses **904**, **1004** or VR headset **1150**, **1220**, and their one or more components are described below in reference to FIGS. **15-17**.

[0091] Referring to FIG. **9**, wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** can communicatively couple via a network **925** (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.). Additionally, wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** can also communicatively couple with one or more servers **930**, computers **940** (e.g., laptops, computers, etc.), mobile devices **950** (e.g., smartphones, tablets, etc.), and/or other electronic devices via network **925** (e.g., cellular, near field, Wi-Fi, personal area network, wireless LAN, etc.).

[0092] In FIG. **9**, a user **908** is shown wearing wrist-wearable device **902** and AR glasses **904** and having HIPD **906** on their desk. The wrist-wearable device **902**, AR glasses **904**, and HIPD **906** facilitate user interaction with an AR environment. In particular, as shown by first AR system **900**, wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** cause presentation of one or more avatars **910**, digital representations of contacts **912**, and virtual objects **914**. As discussed below, user **908** can interact with one or more avatars **910**, digital representations of contacts **912**, and virtual objects **914** via wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906**.

[0093] User **908** can use any of wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** to provide user inputs. For example, user **908** can perform one or more hand gestures that are detected by wrist-wearable device **902** (e.g., using one or more EMG sensors and/or IMUs, described below in reference to FIGS. **13** and **14**) and/or AR glasses **904** (e.g., using one or more image sensor or camera, described below in reference to FIGS. **15-10**) to provide a user input. Alternatively, or additionally, user **908** can provide a user input via one or more touch surfaces of wrist-wearable device **902**, AR glasses **904**, HIPD **906**, and/or voice commands captured by a microphone of wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906**. In some embodiments, wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** include a digital assistant to help user **908** in providing a user input (e.g., completing a sequence of operations, suggesting different operations or commands, providing reminders, confirming a command, etc.). In some embodiments, user **908** can provide a user input via one or more facial gestures and/or facial expressions. For example, cameras of wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** can track eyes of user **908**

for navigating a user interface.

[0094] Wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** can operate alone or in conjunction to allow user **908** to interact with the AR environment. In some embodiments, HIPD **906** is configured to operate as a central hub or control center for the wrist-wearable device **902**, AR glasses **904**, and/or another communicatively coupled device. For example, user **908** can provide an input to interact with the AR environment at any of wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906**, and HIPD **906** can identify one or more back-end and front-end tasks to cause the performance of the requested interaction and distribute instructions to cause the performance of the one or more back-end and front-end tasks at wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906**. In some embodiments, a back-end task is a background processing task that is not perceptible by the user (e.g., rendering content, decompression, compression, etc.), and a front-end task is a user-facing task that is perceptible to the user (e.g., presenting information to the user, providing feedback to the user, etc.). HIPD **906** can perform the back-end tasks and provide wrist-wearable device **902** and/or AR glasses **904** operational data corresponding to the performed back-end tasks such that wrist-wearable device **902** and/or AR glasses **904** can perform the front-end tasks. In this way, HIPD **906**, which has more computational resources and greater thermal headroom than wrist-wearable device **902** and/or AR glasses **904**, performs computationally intensive tasks and reduces the computer resource utilization and/or power usage of wrist-wearable device **902** and/or AR glasses **904**.

[0095] In the example shown by first AR system **900**, HIPD **906** identifies one or more back-end tasks and front-end tasks associated with a user request to initiate an AR video call with one or more other users (represented by avatar **910** and the digital representation of contact **912**) and distributes instructions to cause the performance of the one or more back-end tasks and front-end tasks. In particular, HIPD **906** performs back-end tasks for processing and/or rendering image data (and other data) associated with the AR video call and provides operational data associated with the performed back-end tasks to AR glasses **904** such that the AR glasses **904** perform front-end tasks for presenting the AR video call (e.g., presenting avatar **910** and digital representation of contact **912**).

[0096] In some embodiments, HIPD **906** can operate as a focal or anchor point for causing the presentation of information. This allows user **908** to be generally aware of where information is presented. For example, as shown in first AR system **900**, avatar **910** and the digital representation of contact **912** are presented above HIPD **906**. In particular, HIPD **906** and AR glasses **904** operate in conjunction to determine a location for presenting avatar **910** and the digital representation of contact **912**. In some embodiments, information can be presented a predetermined distance from HIPD **906** (e.g., within 5 meters). For example, as shown in first AR system **900**, virtual object **914** is presented on the desk some distance from HIPD **906**. Similar to the above example, HIPD **906** and AR glasses **904** can operate in conjunction to determine a location for presenting virtual object **914**. Alternatively, in some embodiments, presentation of information is not bound by HIPD **906**. More specifically, avatar **910**, digital representation of contact **912**, and virtual object **914** do not have to be presented within a predetermined distance of HIPD **906**.

[0097] User inputs provided at wrist-wearable device **902**, AR glasses **904**, and/or HIPD **906** are coordinated such that the user can use any device to initiate, continue, and/or complete an operation. For example, user **908** can provide a user input to AR glasses **904** to cause AR glasses **904** to present virtual object **914** and, while virtual object **914** is presented by AR glasses **904**, user **908** can provide one or more hand gestures via wrist-wearable device **902** to interact and/or manipulate virtual object **914**.

[0098] FIG. **10** shows a user **1008** wearing a wrist-wearable device **1002** and AR glasses **1004**, and holding an HIPD **1006**. In second AR system **1000**, the wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006** are used to receive and/or provide one or more messages to a contact of user **1008**. In particular, wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006** detect

and coordinate one or more user inputs to initiate a messaging application and prepare a response to a received message via the messaging application.

[0099] In some embodiments, user **1008** initiates, via a user input, an application on wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006** that causes the application to initiate on at least one device. For example, in second AR system **1000**, user **1008** performs a hand gesture associated with a command for initiating a messaging application (represented by messaging user interface **1016**), wrist-wearable device **1002** detects the hand gesture and, based on a determination that user **1008** is wearing AR glasses **1004**, causes AR glasses **1004** to present a messaging user interface **1016** of the messaging application. AR glasses **1004** can present messaging user interface **1016** to user **1008** via its display (e.g., as shown by a field of view **1018** of user **1008**). In some embodiments, the application is initiated and executed on the device (e.g., wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006**) that detects the user input to initiate the application, and the device provides another device operational data to cause the presentation of the messaging application. For example, wrist-wearable device **1002** can detect the user input to initiate a messaging application, initiate and run the messaging application, and provide operational data to AR glasses **1004** and/or HIPD **1006** to cause presentation of the messaging application. Alternatively, the application can be initiated and executed at a device other than the device that detected the user input. For example, wrist-wearable device **1002** can detect the hand gesture associated with initiating the messaging application and cause HIPD **1006** to run the messaging application and coordinate the presentation of the messaging application.

[0100] Further, user **1008** can provide a user input provided at wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006** to continue and/or complete an operation initiated at another device. For example, after initiating the messaging application via wrist-wearable device **1002** and while AR glasses **1004** present messaging user interface **1016**, user **1008** can provide an input at HIPD **1006** to prepare a response (e.g., shown by the swipe gesture performed on HIPD **1006**). Gestures performed by user **1008** on HIPD **1006** can be provided and/or displayed on another device. For example, a swipe gesture performed on HIPD **1006** is displayed on a virtual keyboard of messaging user interface **1016** displayed by AR glasses **1004**.

[0101] In some embodiments, wrist-wearable device **1002**, AR glasses **1004**, HIPD **1006**, and/or any other communicatively coupled device can present one or more notifications to user **1008**. The notification can be an indication of a new message, an incoming call, an application update, a status update, etc. User **1008** can select the notification via wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006** and can cause presentation of an application or operation associated with the notification on at least one device. For example, user **1008** can receive a notification that a message was received at wrist-wearable device **1002**, AR glasses **1004**, HIPD **1006**, and/or any other communicatively coupled device and can then provide a user input at wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006** to review the notification, and the device detecting the user input can cause an application associated with the notification to be initiated and/or presented at wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006**.

[0102] While the above example describes coordinated inputs used to interact with a messaging application, user inputs can be coordinated to interact with any number of applications including, but not limited to, gaming applications, social media applications, camera applications, web-based applications, financial applications, etc. For example, AR glasses **1004** can present to user **1008** game application data, and HIPD **1006** can be used as a controller to provide inputs to the game. Similarly, user **1008** can use wrist-wearable device **1002** to initiate a camera of AR glasses **1004**, and user **1008** can use wrist-wearable device **1002**, AR glasses **1004**, and/or HIPD **1006** to manipulate the image capture (e.g., zoom in or out, apply filters, etc.) and capture image data.

[0103] Users may interact with the devices disclosed herein in a variety of ways. For example, as shown in FIGS. **11A** and **11B**, a user **1108** may interact with an AR system **1100** by donning a VR headset **1150** while holding HIPD **1106** and wearing wrist-wearable device **1102**. In this example,

AR system **1100** may enable a user to interact with a game **1110** by swiping their arm. One or more of VR headset **1150**, HIPD **1106**, and wrist-wearable device **1102** may detect this gesture and, in response, may display a sword strike in game **1110**. Similarly, in FIGS. **12A** and **12B**, a user **1208** may interact with an AR system **1200** by donning a VR headset **1220** while wearing haptic device **1260** and wrist-wearable device **1230**. In this example, AR system **1200** may enable a user to interact with a game **1210** by swiping their arm. One or more of VR headset **1220**, haptic device **1260**, and wrist-wearable device **1230** may detect this gesture and, in response, may display a spell being cast in game **1110**.

[0104] Having discussed example AR systems, devices for interacting with such AR systems and other computing systems more generally will now be discussed in greater detail. Some explanations of devices and components that can be included in some or all of the example devices discussed below are explained herein for ease of reference. Certain types of the components described below may be more suitable for a particular set of devices, and less suitable for a different set of devices. But subsequent reference to the components explained here should be considered to be encompassed by the descriptions provided.

[0105] In some embodiments discussed below, example devices and systems, including electronic devices and systems, will be addressed. Such example devices and systems are not intended to be limiting, and one of skill in the art will understand that alternative devices and systems to the example devices and systems described herein may be used to perform the operations and construct the systems and devices that are described herein.

[0106] An electronic device may be a device that uses electrical energy to perform a specific function. An electronic device can be any physical object that contains electronic components such as transistors, resistors, capacitors, diodes, and integrated circuits. Examples of electronic devices include smartphones, laptops, digital cameras, televisions, gaming consoles, and music players, as well as the example electronic devices discussed herein. As described herein, an intermediary electronic device may be a device that sits between two other electronic devices and/or a subset of components of one or more electronic devices and facilitates communication, data processing, and/or data transfer between the respective electronic devices and/or electronic components.

[0107] An integrated circuit may be an electronic device made up of multiple interconnected electronic components such as transistors, resistors, and capacitors. These components may be etched onto a small piece of semiconductor material, such as silicon. Integrated circuits may include analog integrated circuits, digital integrated circuits, mixed signal integrated circuits, and/or any other suitable type or form of integrated circuit. Examples of integrated circuits include application-specific integrated circuits (ASICs), processing units, central processing units (CPUs), co-processors, and accelerators.

[0108] Analog integrated circuits, such as sensors, power management circuits, and operational amplifiers, may process continuous signals and perform analog functions such as amplification, active filtering, demodulation, and mixing. Examples of analog integrated circuits include linear integrated circuits and radio frequency circuits.

[0109] Digital integrated circuits, which may be referred to as logic integrated circuits, may include microprocessors, microcontrollers, memory chips, interfaces, power management circuits, programmable devices, and/or any other suitable type or form of integrated circuit. In some embodiments, examples of integrated circuits include central processing units (CPUs),

[0110] Processing units, such as CPUs, may be electronic components that are responsible for executing instructions and controlling the operation of an electronic device (e.g., a computer). There are various types of processors that may be used interchangeably, or may be specifically required, by embodiments described herein. For example, a processor may be: (i) a general processor designed to perform a wide range of tasks, such as running software applications, managing operating systems, and performing arithmetic and logical operations; (ii) a microcontroller designed for specific tasks such as controlling electronic devices, sensors, and

motors; (iii) an accelerator, such as a graphics processing unit (GPU), designed to accelerate the creation and rendering of images, videos, and animations (e.g., virtual-reality animations, such as three-dimensional modeling); (iv) a field-programmable gate array (FPGA) that can be programmed and reconfigured after manufacturing and/or can be customized to perform specific tasks, such as signal processing, cryptography, and machine learning; and/or (v) a digital signal processor (DSP) designed to perform mathematical operations on signals such as audio, video, and radio waves. One or more processors of one or more electronic devices may be used in various embodiments described herein.

[0111] Memory generally refers to electronic components in a computer or electronic device that store data and instructions for the processor to access and manipulate. Examples of memory can include: (i) random access memory (RAM) configured to store data and instructions temporarily; (ii) read-only memory (ROM) configured to store data and instructions permanently (e.g., one or more portions of system firmware, and/or boot loaders) and/or semi-permanently; (iii) flash memory, which can be configured to store data in electronic devices (e.g., USB drives, memory cards, and/or solid-state drives (SSDs)); and/or (iv) cache memory configured to temporarily store frequently accessed data and instructions. Memory, as described herein, can store structured data (e.g., SQL databases, MongoDB databases, GraphQL data, JSON data, etc.). Other examples of data stored in memory can include (i) profile data, including user account data, user settings, and/or other user data stored by the user, (ii) sensor data detected and/or otherwise obtained by one or more sensors, (iii) media content data including stored image data, audio data, documents, and the like, (iv) application data, which can include data collected and/or otherwise obtained and stored during use of an application, and/or any other types of data described herein.

[0112] Controllers may be electronic components that manage and coordinate the operation of other components within an electronic device (e.g., controlling inputs, processing data, and/or generating outputs). Examples of controllers can include: (i) microcontrollers, including small, low-power controllers that are commonly used in embedded systems and Internet of Things (IoT) devices; (ii) programmable logic controllers (PLCs) that may be configured to be used in industrial automation systems to control and monitor manufacturing processes; (iii) system-on-a-chip (SoC) controllers that integrate multiple components such as processors, memory, I/O interfaces, and other peripherals into a single chip; and/or (iv) DSPs.

[0113] A power system of an electronic device may be configured to convert incoming electrical power into a form that can be used to operate the device. A power system can include various components, such as (i) a power source, which can be an alternating current (AC) adapter or a direct current (DC) adapter power supply, (ii) a charger input, which can be configured to use a wired and/or wireless connection (which may be part of a peripheral interface, such as a USB, micro-USB interface, near-field magnetic coupling, magnetic inductive and magnetic resonance charging, and/or radio frequency (RF) charging), (iii) a power-management integrated circuit, configured to distribute power to various components of the device and to ensure that the device operates within safe limits (e.g., regulating voltage, controlling current flow, and/or managing heat dissipation), and/or (iv) a battery configured to store power to provide usable power to components of one or more electronic devices.

[0114] Peripheral interfaces may be electronic components (e.g., of electronic devices) that allow electronic devices to communicate with other devices or peripherals and can provide the ability to input and output data and signals. Examples of peripheral interfaces can include (i) universal serial bus (USB) and/or micro-USB interfaces configured for connecting devices to an electronic device, (ii) Bluetooth interfaces configured to allow devices to communicate with each other, including Bluetooth low energy (BLE), (iii) near field communication (NFC) interfaces configured to be short-range wireless interfaces for operations such as access control, (iv) POGO pins, which may be small, spring-loaded pins configured to provide a charging interface, (v) wireless charging interfaces, (vi) GPS interfaces, (vii) Wi-Fi interfaces for providing a connection between a device

and a wireless network, and/or (viii) sensor interfaces.

[0115] Sensors may be electronic components (e.g., in and/or otherwise in electronic communication with electronic devices, such as wearable devices) configured to detect physical and environmental changes and generate electrical signals. Examples of sensors can include (i) imaging sensors for collecting imaging data (e.g., including one or more cameras disposed on a respective electronic device), (ii) biopotential-signal sensors, (iii) inertial measurement units (e.g., IMUs) for detecting, for example, angular rate, force, magnetic field, and/or changes in acceleration, (iv) heart rate sensors for measuring a user's heart rate, (v) SpO2 sensors for measuring blood oxygen saturation and/or other biometric data of a user, (vi) capacitive sensors for detecting changes in potential at a portion of a user's body (e.g., a sensor-skin interface), and/or (vii) light sensors (e.g., time-of-flight sensors, infrared light sensors, visible light sensors, etc.).

[0116] Biopotential-signal-sensing components may be devices used to measure electrical activity within the body (e.g., biopotential-signal sensors). Some types of biopotential-signal sensors include (i) electroencephalography (EEG) sensors configured to measure electrical activity in the brain to diagnose neurological disorders, (ii) electrocardiogram sensors configured to measure electrical activity of the heart to diagnose heart problems, (iii) electromyography (EMG) sensors configured to measure the electrical activity of muscles and to diagnose neuromuscular disorders, and (iv) electrooculography (EOG) sensors configured to measure the electrical activity of eye muscles to detect eye movement and diagnose eye disorders.

[0117] An application stored in memory of an electronic device (e.g., software) may include instructions stored in the memory. Examples of such applications include (i) games, (ii) word processors, (iii) messaging applications, (iv) media-streaming applications, (v) financial applications, (vi) calendars, (vii) clocks, and (viii) communication interface modules for enabling wired and/or wireless connections between different respective electronic devices (e.g., IEEE 1502.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, or MiWi), custom or standard wired protocols (e.g., Ethernet or HomePlug), and/or any other suitable communication protocols).

[0118] A communication interface may be a mechanism that enables different systems or devices to exchange information and data with each other, including hardware, software, or a combination of both hardware and software. For example, a communication interface can refer to a physical connector and/or port on a device that enables communication with other devices (e.g., USB, Ethernet, HDMI, Bluetooth). In some embodiments, a communication interface can refer to a software layer that enables different software programs to communicate with each other (e.g., application programming interfaces (APIs), protocols like HTTP and TCP/IP, etc.).

[0119] A graphics module may be a component or software module that is designed to handle graphical operations and/or processes and can include a hardware module and/or a software module.

[0120] Non-transitory computer-readable storage media may be physical devices or storage media that can be used to store electronic data in a non-transitory form (e.g., such that the data is stored permanently until it is intentionally deleted or modified).

[0121] FIGS. 13 and 14 illustrate an example wrist-wearable device 1300 and an example computer system 1400, in accordance with some embodiments. Wrist-wearable device 1300 is an instance of wearable device 902 described in FIG. 9 herein, such that the wearable device 902 should be understood to have the features of the wrist-wearable device 1300 and vice versa. FIG. 14 illustrates components of the wrist-wearable device 1300, which can be used individually or in combination, including combinations that include other electronic devices and/or electronic components.

[0122] FIG. 13 shows a wearable band 1310 and a watch body 1320 (or capsule) being coupled, as discussed below, to form wrist-wearable device 1300. Wrist-wearable device 1300 can perform various functions and/or operations associated with navigating through user interfaces and

selectively opening applications as well as the functions and/or operations described above with reference to FIGS. **9-12B**.

[0123] As will be described in more detail below, operations executed by wrist-wearable device **1300** can include (i) presenting content to a user (e.g., displaying visual content via a display **1305**), (ii) detecting (e.g., sensing) user input (e.g., sensing a touch on peripheral button **1323** and/or at a touch screen of the display **1305**, a hand gesture detected by sensors (e.g., biopotential sensors)), (iii) sensing biometric data (e.g., neuromuscular signals, heart rate, temperature, sleep, etc.) via one or more sensors **1313**, messaging (e.g., text, speech, video, etc.); image capture via one or more imaging devices or cameras **1325**, wireless communications (e.g., cellular, near field, Wi-Fi, personal area network, etc.), location determination, financial transactions, providing haptic feedback, providing alarms, providing notifications, providing biometric authentication, providing health monitoring, providing sleep monitoring, etc.

[0124] The above-example functions can be executed independently in watch body **1320**, independently in wearable band **1310**, and/or via an electronic communication between watch body **1320** and wearable band **1310**. In some embodiments, functions can be executed on wrist-wearable device **1300** while an AR environment is being presented (e.g., via one of AR systems **900** to **1200**). The wearable devices described herein can also be used with other types of AR environments.

[0125] Wearable band **1310** can be configured to be worn by a user such that an inner surface of a wearable structure **1311** of wearable band **1310** is in contact with the user's skin. In this example, when worn by a user, sensors **1313** may contact the user's skin. In some examples, one or more of sensors **1313** can sense biometric data such as a user's heart rate, a saturated oxygen level, temperature, sweat level, neuromuscular signals, or a combination thereof. One or more of sensors **1313** can also sense data about a user's environment including a user's motion, altitude, location, orientation, gait, acceleration, position, or a combination thereof. In some embodiment, one or more of sensors **1313** can be configured to track a position and/or motion of wearable band **1310**. One or more of sensors **1313** can include any of the sensors defined above and/or discussed below with respect to FIG. **13**.

[0126] One or more of sensors **1313** can be distributed on an inside and/or an outside surface of wearable band **1310**. In some embodiments, one or more of sensors **1313** are uniformly spaced along wearable band **1310**. Alternatively, in some embodiments, one or more of sensors **1313** are positioned at distinct points along wearable band **1310**. As shown in FIG. **13**, one or more of sensors **1313** can be the same or distinct. For example, in some embodiments, one or more of sensors **1313** can be shaped as a pill (e.g., sensor **1313a**), an oval, a circle a square, an oblong (e.g., sensor **1313c**) and/or any other shape that maintains contact with the user's skin (e.g., such that neuromuscular signal and/or other biometric data can be accurately measured at the user's skin). In some embodiments, one or more sensors of **1313** are aligned to form pairs of sensors (e.g., for sensing neuromuscular signals based on differential sensing within each respective sensor). For example, sensor **1313b** may be aligned with an adjacent sensor to form sensor pair **1314a** and sensor **1313d** may be aligned with an adjacent sensor to form sensor pair **1314b**. In some embodiments, wearable band **1310** does not have a sensor pair. Alternatively, in some embodiments, wearable band **1310** has a predetermined number of sensor pairs (one pair of sensors, three pairs of sensors, four pairs of sensors, six pairs of sensors, sixteen pairs of sensors, etc.).

[0127] Wearable band **1310** can include any suitable number of sensors **1313**. In some embodiments, the number and arrangement of sensors **1313** depends on the particular application for which wearable band **1310** is used. For instance, wearable band **1310** can be configured as an armband, wristband, or chest-band that include a plurality of sensors **1313** with different number of sensors **1313**, a variety of types of individual sensors with the plurality of sensors **1313**, and different arrangements for each use case, such as medical use cases as compared to gaming or

general day-to-day use cases.

[0128] In accordance with some embodiments, wearable band **1310** further includes an electrical ground electrode and a shielding electrode. The electrical ground and shielding electrodes, like the sensors **1313**, can be distributed on the inside surface of the wearable band **1310** such that they contact a portion of the user's skin. For example, the electrical ground and shielding electrodes can be at an inside surface of a coupling mechanism **1316** or an inside surface of a wearable structure **1311**. The electrical ground and shielding electrodes can be formed and/or use the same components as sensors **1313**. In some embodiments, wearable band **1310** includes more than one electrical ground electrode and more than one shielding electrode.

[0129] Sensors **1313** can be formed as part of wearable structure **1311** of wearable band **1310**. In some embodiments, sensors **1313** are flush or substantially flush with wearable structure **1311** such that they do not extend beyond the surface of wearable structure **1311**. While flush with wearable structure **1311**, sensors **1313** are still configured to contact the user's skin (e.g., via a skin-contacting surface). Alternatively, in some embodiments, sensors **1313** extend beyond wearable structure **1311** a predetermined distance (e.g., 0.1-2 mm) to make contact and depress into the user's skin. In some embodiment, sensors **1313** are coupled to an actuator (not shown) configured to adjust an extension height (e.g., a distance from the surface of wearable structure **1311**) of sensors **1313** such that sensors **1313** make contact and depress into the user's skin. In some embodiments, the actuators adjust the extension height between 0.01 mm-1.2 mm. This may allow a the user to customize the positioning of sensors **1313** to improve the overall comfort of the wearable band **1310** when worn while still allowing sensors **1313** to contact the user's skin. In some embodiments, sensors **1313** are indistinguishable from wearable structure **1311** when worn by the user.

[0130] Wearable structure **1311** can be formed of an elastic material, elastomers, etc., configured to be stretched and fitted to be worn by the user. In some embodiments, wearable structure **1311** is a textile or woven fabric. As described above, sensors **1313** can be formed as part of a wearable structure **1311**. For example, sensors **1313** can be molded into the wearable structure **1311**, be integrated into a woven fabric (e.g., sensors **1313** can be sewn into the fabric and mimic the pliability of fabric and can and/or be constructed from a series woven strands of fabric).

[0131] Wearable structure **1311** can include flexible electronic connectors that interconnect sensors **1313**, the electronic circuitry, and/or other electronic components (described below in reference to FIG. **14**) that are enclosed in wearable band **1310**. In some embodiments, the flexible electronic connectors are configured to interconnect sensors **1313**, the electronic circuitry, and/or other electronic components of wearable band **1310** with respective sensors and/or other electronic components of another electronic device (e.g., watch body **1320**). The flexible electronic connectors are configured to move with wearable structure **1311** such that the user adjustment to wearable structure **1311** (e.g., resizing, pulling, folding, etc.) does not stress or strain the electrical coupling of components of wearable band **1310**.

[0132] As described above, wearable band **1310** is configured to be worn by a user. In particular, wearable band **1310** can be shaped or otherwise manipulated to be worn by a user. For example, wearable band **1310** can be shaped to have a substantially circular shape such that it can be configured to be worn on the user's lower arm or wrist. Alternatively, wearable band **1310** can be shaped to be worn on another body part of the user, such as the user's upper arm (e.g., around a bicep), forearm, chest, legs, etc. Wearable band **1310** can include a retaining mechanism **1312** (e.g., a buckle, a hook and loop fastener, etc.) for securing wearable band **1310** to the user's wrist or other body part. While wearable band **1310** is worn by the user, sensors **1313** sense data (referred to as sensor data) from the user's skin. In some examples, sensors **1313** of wearable band **1310** obtain (e.g., sense and record) neuromuscular signals.

[0133] The sensed data (e.g., sensed neuromuscular signals) can be used to detect and/or determine the user's intention to perform certain motor actions. In some examples, sensors **1313** may sense

and record neuromuscular signals from the user as the user performs muscular activations (e.g., movements, gestures, etc.). The detected and/or determined motor actions (e.g., phalange (or digit) movements, wrist movements, hand movements, and/or other muscle intentions) can be used to determine control commands or control information (instructions to perform certain commands after the data is sensed) for causing a computing device to perform one or more input commands. For example, the sensed neuromuscular signals can be used to control certain user interfaces displayed on display **1305** of wrist-wearable device **1300** and/or can be transmitted to a device responsible for rendering an artificial-reality environment (e.g., a head-mounted display) to perform an action in an associated artificial-reality environment, such as to control the motion of a virtual device displayed to the user. The muscular activations performed by the user can include static gestures, such as placing the user's hand palm down on a table, dynamic gestures, such as grasping a physical or virtual object, and covert gestures that are imperceptible to another person, such as slightly tensing a joint by co-contracting opposing muscles or using sub-muscular activations. The muscular activations performed by the user can include symbolic gestures (e.g., gestures mapped to other gestures, interactions, or commands, for example, based on a gesture vocabulary that specifies the mapping of gestures to commands).

[0134] The sensor data sensed by sensors **1313** can be used to provide a user with an enhanced interaction with a physical object (e.g., devices communicatively coupled with wearable band **1310**) and/or a virtual object in an artificial-reality application generated by an artificial-reality system (e.g., user interface objects presented on the display **1305**, or another computing device (e.g., a smartphone)).

[0135] In some embodiments, wearable band **1310** includes one or more haptic devices **1446** (e.g., a vibratory haptic actuator) that are configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user's skin. Sensors **1313** and/or haptic devices **1446** (shown in FIG. **14**) can be configured to operate in conjunction with multiple applications including, without limitation, health monitoring, social media, games, and artificial reality (e.g., the applications associated with artificial reality).

[0136] Wearable band **1310** can also include coupling mechanism **1316** for detachably coupling a capsule (e.g., a computing unit) or watch body **1320** (via a coupling surface of the watch body **1320**) to wearable band **1310**. For example, a cradle or a shape of coupling mechanism **1316** can correspond to shape of watch body **1320** of wrist-wearable device **1300**. In particular, coupling mechanism **1316** can be configured to receive a coupling surface proximate to the bottom side of watch body **1320** (e.g., a side opposite to a front side of watch body **1320** where display **1305** is located), such that a user can push watch body **1320** downward into coupling mechanism **1316** to attach watch body **1320** to coupling mechanism **1316**. In some embodiments, coupling mechanism **1316** can be configured to receive a top side of the watch body **1320** (e.g., a side proximate to the front side of watch body **1320** where display **1305** is located) that is pushed upward into the cradle, as opposed to being pushed downward into coupling mechanism **1316**. In some embodiments, coupling mechanism **1316** is an integrated component of wearable band **1310** such that wearable band **1310** and coupling mechanism **1316** are a single unitary structure. In some embodiments, coupling mechanism **1316** is a type of frame or shell that allows watch body **1320** coupling surface to be retained within or on wearable band **1310** coupling mechanism **1316** (e.g., a cradle, a tracker band, a support base, a clasp, etc.).

[0137] Coupling mechanism **1316** can allow for watch body **1320** to be detachably coupled to the wearable band **1310** through a friction fit, magnetic coupling, a rotation-based connector, a shear-pin coupler, a retention spring, one or more magnets, a clip, a pin shaft, a hook and loop fastener, or a combination thereof. A user can perform any type of motion to couple the watch body **1320** to wearable band **1310** and to decouple the watch body **1320** from the wearable band **1310**. For example, a user can twist, slide, turn, push, pull, or rotate watch body **1320** relative to wearable band **1310**, or a combination thereof, to attach watch body **1320** to wearable band **1310** and to

detach watch body **1320** from wearable band **1310**. Alternatively, as discussed below, in some embodiments, the watch body **1320** can be decoupled from the wearable band **1310** by actuation of a release mechanism **1329**.

[0138] Wearable band **1310** can be coupled with watch body **1320** to increase the functionality of wearable band **1310** (e.g., converting wearable band **1310** into wrist-wearable device **1300**, adding an additional computing unit and/or battery to increase computational resources and/or a battery life of wearable band **1310**, adding additional sensors to improve sensed data, etc.). As described above, wearable band **1310** and coupling mechanism **1316** are configured to operate independently (e.g., execute functions independently) from watch body **1320**. For example, coupling mechanism **1316** can include one or more sensors **1313** that contact a user's skin when wearable band **1310** is worn by the user, with or without watch body **1320** and can provide sensor data for determining control commands.

[0139] A user can detach watch body **1320** from wearable band **1310** to reduce the encumbrance of wrist-wearable device **1300** to the user. For embodiments in which watch body **1320** is removable, watch body **1320** can be referred to as a removable structure, such that in these embodiments wrist-wearable device **1300** includes a wearable portion (e.g., wearable band **1310**) and a removable structure (e.g., watch body **1320**).

[0140] Turning to watch body **1320**, in some examples watch body **1320** can have a substantially rectangular or circular shape. Watch body **1320** is configured to be worn by the user on their wrist or on another body part. More specifically, watch body **1320** is sized to be easily carried by the user, attached on a portion of the user's clothing, and/or coupled to wearable band **1310** (forming the wrist-wearable device **1300**). As described above, watch body **1320** can have a shape corresponding to coupling mechanism **1316** of wearable band **1310**. In some embodiments, watch body **1320** includes a single release mechanism **1329** or multiple release mechanisms (e.g., two release mechanisms **1329** positioned on opposing sides of watch body **1320**, such as spring-loaded buttons) for decoupling watch body **1320** from wearable band **1310**. Release mechanism **1329** can include, without limitation, a button, a knob, a plunger, a handle, a lever, a fastener, a clasp, a dial, a latch, or a combination thereof.

[0141] A user can actuate release mechanism **1329** by pushing, turning, lifting, depressing, shifting, or performing other actions on release mechanism **1329**. Actuation of release mechanism **1329** can release (e.g., decouple) watch body **1320** from coupling mechanism **1316** of wearable band **1310**, allowing the user to use watch body **1320** independently from wearable band **1310** and vice versa. For example, decoupling watch body **1320** from wearable band **1310** can allow a user to capture images using rear-facing camera **1325b**. Although release mechanism **1329** is shown positioned at a corner of watch body **1320**, release mechanism **1329** can be positioned anywhere on watch body **1320** that is convenient for the user to actuate. In addition, in some embodiments, wearable band **1310** can also include a respective release mechanism for decoupling watch body **1320** from coupling mechanism **1316**. In some embodiments, release mechanism **1329** is optional and watch body **1320** can be decoupled from coupling mechanism **1316** as described above (e.g., via twisting, rotating, etc.).

[0142] Watch body **1320** can include one or more peripheral buttons **1323** and **1327** for performing various operations at watch body **1320**. For example, peripheral buttons **1323** and **1327** can be used to turn on or wake (e.g., transition from a sleep state to an active state) display **1305**, unlock watch body **1320**, increase or decrease a volume, increase or decrease a brightness, interact with one or more applications, interact with one or more user interfaces, etc. Additionally or alternatively, in some embodiments, display **1305** operates as a touch screen and allows the user to provide one or more inputs for interacting with watch body **1320**.

[0143] In some embodiments, watch body **1320** includes one or more sensors **1321**. Sensors **1321** of watch body **1320** can be the same or distinct from sensors **1313** of wearable band **1310**. Sensors **1321** of watch body **1320** can be distributed on an inside and/or an outside surface of watch body

1320. In some embodiments, sensors **1321** are configured to contact a user's skin when watch body **1320** is worn by the user. For example, sensors **1321** can be placed on the bottom side of watch body **1320** and coupling mechanism **1316** can be a cradle with an opening that allows the bottom side of watch body **1320** to directly contact the user's skin. Alternatively, in some embodiments, watch body **1320** does not include sensors that are configured to contact the user's skin (e.g., including sensors internal and/or external to the watch body **1320** that are configured to sense data of watch body **1320** and the surrounding environment). In some embodiments, sensors **1321** are configured to track a position and/or motion of watch body **1320**.

[0144] Watch body **1320** and wearable band **1310** can share data using a wired communication method (e.g., a Universal Asynchronous Receiver/Transmitter (UART), a USB transceiver, etc.) and/or a wireless communication method (e.g., near field communication, Bluetooth, etc.). For example, watch body **1320** and wearable band **1310** can share data sensed by sensors **1313** and **1321**, as well as application and device specific information (e.g., active and/or available applications, output devices (e.g., displays, speakers, etc.), input devices (e.g., touch screens, microphones, imaging sensors, etc.).

[0145] In some embodiments, watch body **1320** can include, without limitation, a front-facing camera **1325a** and/or a rear-facing camera **1325b**, sensors **1321** (e.g., a biometric sensor, an IMU, a heart rate sensor, a saturated oxygen sensor, a neuromuscular signal sensor, an altimeter sensor, a temperature sensor, a bioimpedance sensor, a pedometer sensor, an optical sensor (e.g., imaging sensor **1463**), a touch sensor, a sweat sensor, etc.). In some embodiments, watch body **1320** can include one or more haptic devices **1476** (e.g., a vibratory haptic actuator) that is configured to provide haptic feedback (e.g., a cutaneous and/or kinesthetic sensation, etc.) to the user. Sensors **1421** and/or haptic device **1476** can also be configured to operate in conjunction with multiple applications including, without limitation, health monitoring applications, social media applications, game applications, and artificial reality applications (e.g., the applications associated with artificial reality).

[0146] As described above, watch body **1320** and wearable band **1310**, when coupled, can form wrist-wearable device **1300**. When coupled, watch body **1320** and wearable band **1310** may operate as a single device to execute functions (operations, detections, communications, etc.) described herein. In some embodiments, each device may be provided with particular instructions for performing the one or more operations of wrist-wearable device **1300**. For example, in accordance with a determination that watch body **1320** does not include neuromuscular signal sensors, wearable band **1310** can include alternative instructions for performing associated instructions (e.g., providing sensed neuromuscular signal data to watch body **1320** via a different electronic device). Operations of wrist-wearable device **1300** can be performed by watch body **1320** alone or in conjunction with wearable band **1310** (e.g., via respective processors and/or hardware components) and vice versa. In some embodiments, operations of wrist-wearable device **1300**, watch body **1320**, and/or wearable band **1310** can be performed in conjunction with one or more processors and/or hardware components.

[0147] As described below with reference to the block diagram of FIG. **14**, wearable band **1310** and/or watch body **1320** can each include independent resources required to independently execute functions. For example, wearable band **1310** and/or watch body **1320** can each include a power source (e.g., a battery), a memory, data storage, a processor (e.g., a central processing unit (CPU)), communications, a light source, and/or input/output devices.

[0148] FIG. **14** shows block diagrams of a computing system **1430** corresponding to wearable band **1310** and a computing system **1460** corresponding to watch body **1320** according to some embodiments. Computing system **1400** of wrist-wearable device **1300** may include a combination of components of wearable band computing system **1430** and watch body computing system **1460**, in accordance with some embodiments.

[0149] Watch body **1320** and/or wearable band **1310** can include one or more components shown in

watch body computing system **1460**. In some embodiments, a single integrated circuit may include all or a substantial portion of the components of watch body computing system **1460** included in a single integrated circuit. Alternatively, in some embodiments, components of the watch body computing system **1460** may be included in a plurality of integrated circuits that are communicatively coupled. In some embodiments, watch body computing system **1460** may be configured to couple (e.g., via a wired or wireless connection) with wearable band computing system **1430**, which may allow the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0150] Watch body computing system **1460** can include one or more processors **1479**, a controller **1477**, a peripherals interface **1461**, a power system **1495**, and memory (e.g., a memory **1480**).

[0151] Power system **1495** can include a charger input **1496**, a power-management integrated circuit (PMIC) **1497**, and a battery **1498**. In some embodiments, a watch body **1320** and a wearable band **1310** can have respective batteries (e.g., battery **1498** and **1459**) and can share power with each other. Watch body **1320** and wearable band **1310** can receive a charge using a variety of techniques. In some embodiments, watch body **1320** and wearable band **1310** can use a wired charging assembly (e.g., power cords) to receive the charge. Alternatively, or in addition, watch body **1320** and/or wearable band **1310** can be configured for wireless charging. For example, a portable charging device can be designed to mate with a portion of watch body **1320** and/or wearable band **1310** and wirelessly deliver usable power to battery **1498** of watch body **1320** and/or battery **1459** of wearable band **1310**. Watch body **1320** and wearable band **1310** can have independent power systems (e.g., power system **1495** and **1456**, respectively) to enable each to operate independently. Watch body **1320** and wearable band **1310** can also share power (e.g., one can charge the other) via respective PMICs (e.g., PMICs **1497** and **1458**) and charger inputs (e.g., **1457** and **1496**) that can share power over power and ground conductors and/or over wireless charging antennas.

[0152] In some embodiments, peripherals interface **1461** can include one or more sensors **1421**. Sensors **1421** can include one or more coupling sensors **1462** for detecting when watch body **1320** is coupled with another electronic device (e.g., a wearable band **1310**). Sensors **1421** can include one or more imaging sensors **1463** (e.g., one or more of cameras **1425**, and/or separate imaging sensors **1463** (e.g., thermal-imaging sensors)). In some embodiments, sensors **1421** can include one or more SpO₂ sensors **1464**. In some embodiments, sensors **1421** can include one or more biopotential-signal sensors (e.g., EMG sensors **1465**, which may be disposed on an interior, user-facing portion of watch body **1320** and/or wearable band **1310**). In some embodiments, sensors **1421** may include one or more capacitive sensors **1466**. In some embodiments, sensors **1421** may include one or more heart rate sensors **1467**. In some embodiments, sensors **1421** may include one or more IMU sensors **1468**. In some embodiments, one or more IMU sensors **1468** can be configured to detect movement of a user's hand or other location where watch body **1320** is placed or held.

[0153] In some embodiments, one or more of sensors **1421** may provide an example human-machine interface. For example, a set of neuromuscular sensors, such as EMG sensors **1465**, may be arranged circumferentially around wearable band **1310** with an interior surface of EMG sensors **1465** being configured to contact a user's skin. Any suitable number of neuromuscular sensors may be used (e.g., between 2 and 20 sensors). The number and arrangement of neuromuscular sensors may depend on the particular application for which the wearable device is used. For example, wearable band **1310** can be used to generate control information for controlling an augmented reality system, a robot, controlling a vehicle, scrolling through text, controlling a virtual avatar, or any other suitable control task.

[0154] In some embodiments, neuromuscular sensors may be coupled together using flexible electronics incorporated into the wireless device, and the output of one or more of the sensing components can be optionally processed using hardware signal processing circuitry (e.g., to

perform amplification, filtering, and/or rectification). In other embodiments, at least some signal processing of the output of the sensing components can be performed in software such as processors **1479**. Thus, signal processing of signals sampled by the sensors can be performed in hardware, software, or by any suitable combination of hardware and software, as aspects of the technology described herein are not limited in this respect.

[0155] Neuromuscular signals may be processed in a variety of ways. For example, the output of EMG sensors **1465** may be provided to an analog front end, which may be configured to perform analog processing (e.g., amplification, noise reduction, filtering, etc.) on the recorded signals. The processed analog signals may then be provided to an analog-to-digital converter, which may convert the analog signals to digital signals that can be processed by one or more computer processors. Furthermore, although this example is as discussed in the context of interfaces with EMG sensors, the embodiments described herein can also be implemented in wearable interfaces with other types of sensors including, but not limited to, mechanomyography (MMG) sensors, sonomyography (SMG) sensors, and electrical impedance tomography (EIT) sensors.

[0156] In some embodiments, peripherals interface **1461** includes a near-field communication (NFC) component **1469**, a global-position system (GPS) component **1470**, a long-term evolution (LTE) component **1471**, and/or a Wi-Fi and/or Bluetooth communication component **1472**. In some embodiments, peripherals interface **1461** includes one or more buttons **1473** (e.g., peripheral buttons **1323** and **1327** in FIG. **13**), which, when selected by a user, cause operation to be performed at watch body **1320**. In some embodiments, the peripherals interface **1461** includes one or more indicators, such as a light emitting diode (LED), to provide a user with visual indicators (e.g., message received, low battery, active microphone and/or camera, etc.).

[0157] Watch body **1320** can include at least one display **1305** for displaying visual representations of information or data to a user, including user-interface elements and/or three-dimensional virtual objects. The display can also include a touch screen for inputting user inputs, such as touch gestures, swipe gestures, and the like. Watch body **1320** can include at least one speaker **1474** and at least one microphone **1475** for providing audio signals to the user and receiving audio input from the user. The user can provide user inputs through microphone **1475** and can also receive audio output from speaker **1474** as part of a haptic event provided by haptic controller **1478**. Watch body **1320** can include at least one camera **1425**, including a front camera **1425a** and a rear camera **1425b**. Cameras **1425** can include ultra-wide-angle cameras, wide angle cameras, fish-eye cameras, spherical cameras, telephoto cameras, depth-sensing cameras, or other types of cameras.

[0158] Watch body computing system **1460** can include one or more haptic controllers **1478** and associated componentry (e.g., haptic devices **1476**) for providing haptic events at watch body **1320** (e.g., a vibrating sensation or audio output in response to an event at the watch body **1320**). Haptic controllers **1478** can communicate with one or more haptic devices **1476**, such as electroacoustic devices, including a speaker of the one or more speakers **1474** and/or other audio components and/or electromechanical devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating components (e.g., a component that converts electrical signals into tactile outputs on the device). Haptic controller **1478** can provide haptic events to that are capable of being sensed by a user of watch body **1320**. In some embodiments, one or more haptic controllers **1478** can receive input signals from an application of applications **1482**.

[0159] In some embodiments, wearable band computing system **1430** and/or watch body computing system **1460** can include memory **1480**, which can be controlled by one or more memory controllers of controllers **1477**. In some embodiments, software components stored in memory **1480** include one or more applications **1482** configured to perform operations at the watch body **1320**. In some embodiments, one or more applications **1482** may include games, word processors, messaging applications, calling applications, web browsers, social media applications, media streaming applications, financial applications, calendars, clocks, etc. In some embodiments,

software components stored in memory **1480** include one or more communication interface modules **1483** as defined above. In some embodiments, software components stored in memory **1480** include one or more graphics modules **1484** for rendering, encoding, and/or decoding audio and/or visual data and one or more data management modules **1485** for collecting, organizing, and/or providing access to data **1487** stored in memory **1480**. In some embodiments, one or more of applications **1482** and/or one or more modules can work in conjunction with one another to perform various tasks at the watch body **1320**.

[0160] In some embodiments, software components stored in memory **1480** can include one or more operating systems **1481** (e.g., a Linux-based operating system, an Android operating system, etc.). Memory **1480** can also include data **1487**. Data **1487** can include profile data **1488A**, sensor data **1489A**, media content data **1490**, and application data **1491**.

[0161] It should be appreciated that watch body computing system **1460** is an example of a computing system within watch body **1320**, and that watch body **1320** can have more or fewer components than shown in watch body computing system **1460**, can combine two or more components, and/or can have a different configuration and/or arrangement of the components. The various components shown in watch body computing system **1460** are implemented in hardware, software, firmware, or a combination thereof, including one or more signal processing and/or application-specific integrated circuits.

[0162] Turning to the wearable band computing system **1430**, one or more components that can be included in wearable band **1310** are shown. Wearable band computing system **1430** can include more or fewer components than shown in watch body computing system **1460**, can combine two or more components, and/or can have a different configuration and/or arrangement of some or all of the components. In some embodiments, all, or a substantial portion of the components of wearable band computing system **1430** are included in a single integrated circuit. Alternatively, in some embodiments, components of wearable band computing system **1430** are included in a plurality of integrated circuits that are communicatively coupled. As described above, in some embodiments, wearable band computing system **1430** is configured to couple (e.g., via a wired or wireless connection) with watch body computing system **1460**, which allows the computing systems to share components, distribute tasks, and/or perform other operations described herein (individually or as a single device).

[0163] Wearable band computing system **1430**, similar to watch body computing system **1460**, can include one or more processors **1449**, one or more controllers **1447** (including one or more haptics controllers **1448**), a peripherals interface **1431** that can include one or more sensors **1413** and other peripheral devices, a power source (e.g., a power system **1456**), and memory (e.g., a memory **1450**) that includes an operating system (e.g., an operating system **1451**), data (e.g., data **1454** including profile data **1488B**, sensor data **1489B**, etc.), and one or more modules (e.g., a communications interface module **1452**, a data management module **1453**, etc.).

[0164] One or more of sensors **1413** can be analogous to sensors **1421** of watch body computing system **1460**. For example, sensors **1413** can include one or more coupling sensors **1432**, one or more SpO₂ sensors **1434**, one or more EMG sensors **1435**, one or more capacitive sensors **1436**, one or more heart rate sensors **1437**, and one or more IMU sensors **1438**.

[0165] Peripherals interface **1431** can also include other components analogous to those included in peripherals interface **1461** of watch body computing system **1460**, including an NFC component **1439**, a GPS component **1440**, an LTE component **1441**, a Wi-Fi and/or Bluetooth communication component **1442**, and/or one or more haptic devices **1446** as described above in reference to peripherals interface **1461**. In some embodiments, peripherals interface **1431** includes one or more buttons **1443**, a display **1433**, a speaker **1444**, a microphone **1445**, and a camera **1455**. In some embodiments, peripherals interface **1431** includes one or more indicators, such as an LED.

[0166] It should be appreciated that wearable band computing system **1430** is an example of a computing system within wearable band **1310**, and that wearable band **1310** can have more or

fewer components than shown in wearable band computing system **1430**, combine two or more components, and/or have a different configuration and/or arrangement of the components. The various components shown in wearable band computing system **1430** can be implemented in one or more of a combination of hardware, software, or firmware, including one or more signal processing and/or application-specific integrated circuits.

[0167] Wrist-wearable device **1300** with respect to FIG. **13** is an example of wearable band **1310** and watch body **1320** coupled together, so wrist-wearable device **1300** will be understood to include the components shown and described for wearable band computing system **1430** and watch body computing system **1460**. In some embodiments, wrist-wearable device **1300** has a split architecture (e.g., a split mechanical architecture, a split electrical architecture, etc.) between watch body **1320** and wearable band **1310**. In other words, all of the components shown in wearable band computing system **1430** and watch body computing system **1460** can be housed or otherwise disposed in a combined wrist-wearable device **1300** or within individual components of watch body **1320**, wearable band **1310**, and/or portions thereof (e.g., a coupling mechanism **1316** of wearable band **1310**).

[0168] The techniques described above can be used with any device for sensing neuromuscular signals but could also be used with other types of wearable devices for sensing neuromuscular signals (such as body-wearable or head-wearable devices that might have neuromuscular sensors closer to the brain or spinal column).

[0169] In some embodiments, wrist-wearable device **1300** can be used in conjunction with a head-wearable device (e.g., AR glasses **1500** and VR system **1610**) and/or an HIPD, and wrist-wearable device **1300** can also be configured to be used to allow a user to control any aspect of the artificial reality (e.g., by using EMG-based gestures to control user interface objects in the artificial reality and/or by allowing a user to interact with the touchscreen on the wrist-wearable device to also control aspects of the artificial reality). Having thus described example wrist-wearable devices, attention will now be turned to example head-wearable devices, such as AR glasses **1500** and VR headset **1610**.

[0170] FIGS. **15** to **17** show example artificial-reality systems, which can be used as or in connection with wrist-wearable device **1300**. In some embodiments, AR system **1500** includes an eyewear device **1502**, as shown in FIG. **15**. In some embodiments, VR system **1610** includes a head-mounted display (HMD) **1612**, as shown in FIGS. **16A** and **16B**. In some embodiments, AR system **1500** and VR system **1610** can include one or more analogous components (e.g., components for presenting interactive artificial-reality environments, such as processors, memory, and/or presentation devices, including one or more displays and/or one or more waveguides), some of which are described in more detail with respect to FIG. **17**. As described herein, a head-wearable device can include components of eyewear device **1502** and/or head-mounted display **1612**. Some embodiments of head-wearable devices do not include any displays, including any of the displays described with respect to AR system **1500** and/or VR system **1610**. While the example artificial-reality systems are respectively described herein as AR system **1500** and VR system **1610**, either or both of the example AR systems described herein can be configured to present fully-immersive virtual-reality scenes presented in substantially all of a user's field of view or subtler augmented-reality scenes that are presented within a portion, less than all, of the user's field of view.

[0171] FIG. **15** show an example visual depiction of AR system **1500**, including an eyewear device **1502** (which may also be described herein as augmented-reality glasses, and/or smart glasses). AR system **1500** can include additional electronic components that are not shown in FIG. **15**, such as a wearable accessory device and/or an intermediary processing device, in electronic communication or otherwise configured to be used in conjunction with the eyewear device **1502**. In some embodiments, the wearable accessory device and/or the intermediary processing device may be configured to couple with eyewear device **1502** via a coupling mechanism in electronic communication with a coupling sensor **1724** (FIG. **17**), where coupling sensor **1724** can detect

when an electronic device becomes physically or electronically coupled with eyewear device **1502**. In some embodiments, eyewear device **1502** can be configured to couple to a housing **1790** (FIG. **17**), which may include one or more additional coupling mechanisms configured to couple with additional accessory devices. The components shown in FIG. **15** can be implemented in hardware, software, firmware, or a combination thereof, including one or more signal-processing components and/or application-specific integrated circuits (ASICs).

[0172] Eyewear device **1502** includes mechanical glasses components, including a frame **1504** configured to hold one or more lenses (e.g., one or both lenses **1506-1** and **1506-2**). One of ordinary skill in the art will appreciate that eyewear device **1502** can include additional mechanical components, such as hinges configured to allow portions of frame **1504** of eyewear device **1502** to be folded and unfolded, a bridge configured to span the gap between lenses **1506-1** and **1506-2** and rest on the user's nose, nose pads configured to rest on the bridge of the nose and provide support for eyewear device **1502**, earpieces configured to rest on the user's ears and provide additional support for eyewear device **1502**, temple arms configured to extend from the hinges to the earpieces of eyewear device **1502**, and the like. One of ordinary skill in the art will further appreciate that some examples of AR system **1500** can include none of the mechanical components described herein. For example, smart contact lenses configured to present artificial reality to users may not include any components of eyewear device **1502**.

[0173] Eyewear device **1502** includes electronic components, many of which will be described in more detail below with respect to FIG. **17**. Some example electronic components are illustrated in FIG. **15**, including acoustic sensors **1525-1**, **1525-2**, **1525-3**, **1525-4**, **1525-5**, and **1525-6**, which can be distributed along a substantial portion of the frame **1504** of eyewear device **1502**. Eyewear device **1502** also includes a left camera **1539A** and a right camera **1539B**, which are located on different sides of the frame **1504**. Eyewear device **1502** also includes a processor **1548** (or any other suitable type or form of integrated circuit) that is embedded into a portion of the frame **1504**.

[0174] FIGS. **16A** and **16B** show a VR system **1610** that includes a head-mounted display (HMD) **1612** (e.g., also referred to herein as an artificial-reality headset, a head-wearable device, a VR headset, etc.), in accordance with some embodiments. As noted, some artificial-reality systems (e.g., AR system **1500**) may, instead of blending an artificial reality with actual reality, substantially replace one or more of a user's visual and/or other sensory perceptions of the real world with a virtual experience (e.g., AR systems **1100** and **1200**).

[0175] HMD **1612** includes a front body **1614** and a frame **1616** (e.g., a strap or band) shaped to fit around a user's head. In some embodiments, front body **1614** and/or frame **1616** include one or more electronic elements for facilitating presentation of and/or interactions with an AR and/or VR system (e.g., displays, IMUs, tracking emitter or detectors). In some embodiments, HMD **1612** includes output audio transducers (e.g., an audio transducer **1618**), as shown in FIG. **16B**. In some embodiments, one or more components, such as the output audio transducer(s) **1618** and frame **1616**, can be configured to attach and detach (e.g., are detachably attachable) to HMD **1612** (e.g., a portion or all of frame **1616**, and/or audio transducer **1618**), as shown in FIG. **16B**. In some embodiments, coupling a detachable component to HMD **1612** causes the detachable component to come into electronic communication with HMD **1612**.

[0176] FIGS. **16A** and **16B** also show that VR system **1610** includes one or more cameras, such as left camera **1639A** and right camera **1639B**, which can be analogous to left and right cameras **1539A** and **1539B** on frame **1504** of eyewear device **1502**. In some embodiments, VR system **1610** includes one or more additional cameras (e.g., cameras **1639C** and **1639D**), which can be configured to augment image data obtained by left and right cameras **1639A** and **1639B** by providing more information. For example, camera **1639C** can be used to supply color information that is not discerned by cameras **1639A** and **1639B**. In some embodiments, one or more of cameras **1639A** to **1639D** can include an optional IR cut filter configured to remove IR light from being received at the respective camera sensors.

[0177] FIG. 17 illustrates a computing system 1720 and an optional housing 1790, each of which show components that can be included in AR system 1500 and/or VR system 1610. In some embodiments, more or fewer components can be included in optional housing 1790 depending on practical restraints of the respective AR system being described.

[0178] In some embodiments, computing system 1720 can include one or more peripherals interfaces 1722A and/or optional housing 1790 can include one or more peripherals interfaces 1722B. Each of computing system 1720 and optional housing 1790 can also include one or more power systems 1742A and 1742B, one or more controllers 1746 (including one or more haptic controllers 1747), one or more processors 1748A and 1748B (as defined above, including any of the examples provided), and memory 1750A and 1750B, which can all be in electronic communication with each other. For example, the one or more processors 1748A and 1748B can be configured to execute instructions stored in memory 1750A and 1750B, which can cause a controller of one or more of controllers 1746 to cause operations to be performed at one or more peripheral devices connected to peripherals interface 1722A and/or 1722B. In some embodiments, each operation described can be powered by electrical power provided by power system 1742A and/or 1742B.

[0179] In some embodiments, peripherals interface 1722A can include one or more devices configured to be part of computing system 1720, some of which have been defined above and/or described with respect to the wrist-wearable devices shown in FIGS. 13 and 14. For example, peripherals interface 1722A can include one or more sensors 1723A. Some example sensors 1723A include one or more coupling sensors 1724, one or more acoustic sensors 1725, one or more imaging sensors 1726, one or more EMG sensors 1727, one or more capacitive sensors 1728, one or more IMU sensors 1729, and/or any other types of sensors explained above or described with respect to any other embodiments discussed herein.

[0180] In some embodiments, peripherals interfaces 1722A and 1722B can include one or more additional peripheral devices, including one or more NFC devices 1730, one or more GPS devices 1731, one or more LTE devices 1732, one or more Wi-Fi and/or Bluetooth devices 1733, one or more buttons 1734 (e.g., including buttons that are slidable or otherwise adjustable), one or more displays 1735A and 1735B, one or more speakers 1736A and 1736B, one or more microphones 1737, one or more cameras 1738A and 1738B (e.g., including the left camera 1739A and/or a right camera 1739B), one or more haptic devices 1740, and/or any other types of peripheral devices defined above or described with respect to any other embodiments discussed herein.

[0181] AR systems can include a variety of types of visual feedback mechanisms (e.g., presentation devices). For example, display devices in AR system 1500 and/or VR system 1610 can include one or more liquid-crystal displays (LCDs), light emitting diode (LED) displays, organic LED (OLED) displays, and/or any other suitable types of display screens. Artificial-reality systems can include a single display screen (e.g., configured to be seen by both eyes), and/or can provide separate display screens for each eye, which can allow for additional flexibility for varifocal adjustments and/or for correcting a refractive error associated with a user's vision. Some embodiments of AR systems also include optical subsystems having one or more lenses (e.g., conventional concave or convex lenses, Fresnel lenses, or adjustable liquid lenses) through which a user can view a display screen.

[0182] For example, respective displays 1735A and 1735B can be coupled to each of the lenses 1506-1 and 1506-2 of AR system 1500. Displays 1735A and 1735B may be coupled to each of lenses 1506-1 and 1506-2, which can act together or independently to present an image or series of images to a user. In some embodiments, AR system 1500 includes a single display 1735A or 1735B (e.g., a near-eye display) or more than two displays 1735A and 1735B. In some embodiments, a first set of one or more displays 1735A and 1735B can be used to present an augmented-reality environment, and a second set of one or more display devices 1735A and 1735B can be used to present a virtual-reality environment. In some embodiments, one or more waveguides are used in conjunction with presenting artificial-reality content to the user of AR system 1500 (e.g., as a

means of delivering light from one or more displays **1735A** and **1735B** to the user's eyes). In some embodiments, one or more waveguides are fully or partially integrated into the eyewear device **1502**. Additionally, or alternatively to display screens, some artificial-reality systems include one or more projection systems. For example, display devices in AR system **1500** and/or VR system **1610** can include micro-LED projectors that project light (e.g., using a waveguide) into display devices, such as clear combiner lenses that allow ambient light to pass through. The display devices can refract the projected light toward a user's pupil and can enable a user to simultaneously view both artificial-reality content and the real world. Artificial-reality systems can also be configured with any other suitable type or form of image projection system. In some embodiments, one or more waveguides are provided additionally or alternatively to the one or more display(s) **1735A** and **1735B**.

[0183] Computing system **1720** and/or optional housing **1790** of AR system **1500** or VR system **1610** can include some or all of the components of a power system **1742A** and **1742B**. Power systems **1742A** and **1742B** can include one or more charger inputs **1743**, one or more PMICs **1744**, and/or one or more batteries **1745A** and **1744B**.

[0184] Memory **1750A** and **1750B** may include instructions and data, some or all of which may be stored as non-transitory computer-readable storage media within the memories **1750A** and **1750B**. For example, memory **1750A** and **1750B** can include one or more operating systems **1751**, one or more applications **1752**, one or more communication interface applications **1753A** and **1753B**, one or more graphics applications **1754A** and **1754B**, one or more AR processing applications **1755A** and **1755B**, and/or any other types of data defined above or described with respect to any other embodiments discussed herein.

[0185] Memory **1750A** and **1750B** also include data **1760A** and **1760B**, which can be used in conjunction with one or more of the applications discussed above. Data **1760A** and **1760B** can include profile data **1761**, sensor data **1762A** and **1762B**, media content data **1763A**, AR application data **1764A** and **1764B**, and/or any other types of data defined above or described with respect to any other embodiments discussed herein.

[0186] In some embodiments, controller **1746** of eyewear device **1502** may process information generated by sensors **1723A** and/or **1723B** on eyewear device **1502** and/or another electronic device within AR system **1500**. For example, controller **1746** can process information from acoustic sensors **1525-1** and **1525-2**. For each detected sound, controller **1746** can perform a direction of arrival (DOA) estimation to estimate a direction from which the detected sound arrived at eyewear device **1502** of AR system **1500**. As one or more of acoustic sensors **1725** (e.g., the acoustic sensors **1525-1**, **1525-2**) detects sounds, controller **1746** can populate an audio data set with the information (e.g., represented in FIG. 17 as sensor data **1762A** and **1762B**).

[0187] In some embodiments, a physical electronic connector can convey information between eyewear device **1502** and another electronic device and/or between one or more processors **1548**, **1748A**, **1748B** of AR system **1500** or VR system **1610** and controller **1746**. The information can be in the form of optical data, electrical data, wireless data, or any other transmittable data form. Moving the processing of information generated by eyewear device **1502** to an intermediary processing device can reduce weight and heat in the eyewear device, making it more comfortable and safer for a user. In some embodiments, an optional wearable accessory device (e.g., an electronic neckband) is coupled to eyewear device **1502** via one or more connectors. The connectors can be wired or wireless connectors and can include electrical and/or non-electrical (e.g., structural) components. In some embodiments, eyewear device **1502** and the wearable accessory device can operate independently without any wired or wireless connection between them.

[0188] In some situations, pairing external devices, such as an intermediary processing device (e.g., HIPD **906**, **1006**, **1106**) with eyewear device **1502** (e.g., as part of AR system **1500**) enables eyewear device **1502** to achieve a similar form factor of a pair of glasses while still providing

sufficient battery and computation power for expanded capabilities. Some, or all, of the battery power, computational resources, and/or additional features of AR system **1500** can be provided by a paired device or shared between a paired device and eyewear device **1502**, thus reducing the weight, heat profile, and form factor of eyewear device **1502** overall while allowing eyewear device **1502** to retain its desired functionality. For example, the wearable accessory device can allow components that would otherwise be included on eyewear device **1502** to be included in the wearable accessory device and/or intermediary processing device, thereby shifting a weight load from the user's head and neck to one or more other portions of the user's body. In some embodiments, the intermediary processing device has a larger surface area over which to diffuse and disperse heat to the ambient environment. Thus, the intermediary processing device can allow for greater battery and computation capacity than might otherwise have been possible on eyewear device **1502** standing alone. Because weight carried in the wearable accessory device can be less invasive to a user than weight carried in the eyewear device **1502**, a user may tolerate wearing a lighter eyewear device and carrying or wearing the paired device for greater lengths of time than the user would tolerate wearing a heavier eyewear device standing alone, thereby enabling an artificial-reality environment to be incorporated more fully into a user's day-to-day activities.

[0189] AR systems can include various types of computer vision components and subsystems. For example, AR system **1500** and/or VR system **1610** can include one or more optical sensors such as two-dimensional (2D) or three-dimensional (3D) cameras, time-of-flight depth sensors, structured light transmitters and detectors, single-beam or sweeping laser rangefinders, 3D LiDAR sensors, and/or any other suitable type or form of optical sensor. An AR system can process data from one or more of these sensors to identify a location of a user and/or aspects of the user's real-world physical surroundings, including the locations of real-world objects within the real-world physical surroundings. In some embodiments, the methods described herein are used to map the real world, to provide a user with context about real-world surroundings, and/or to generate digital twins (e.g., interactable virtual objects), among a variety of other functions. For example, FIGS. **16A** and **16B** show VR system **1610** having cameras **1639A** to **1639D**, which can be used to provide depth information for creating a voxel field and a two-dimensional mesh to provide object information to the user to avoid collisions.

[0190] In some embodiments, AR system **1500** and/or VR system **1610** can include haptic (tactile) feedback systems, which may be incorporated into headwear, gloves, body suits, handheld controllers, environmental devices (e.g., chairs or floormats), and/or any other type of device or system, such as the wearable devices discussed herein. The haptic feedback systems may provide various types of cutaneous feedback, including vibration, force, traction, shear, texture, and/or temperature. The haptic feedback systems may also provide various types of kinesthetic feedback, such as motion and compliance. The haptic feedback may be implemented using motors, piezoelectric actuators, fluidic systems, and/or a variety of other types of feedback mechanisms. The haptic feedback systems may be implemented independently of other artificial-reality devices, within other artificial-reality devices, and/or in conjunction with other artificial-reality devices.

[0191] In some embodiments of an artificial reality system, such as AR system **1500** and/or VR system **1610**, ambient light (e.g., a live feed of the surrounding environment that a user would normally see) can be passed through a display element of a respective head-wearable device presenting aspects of the AR system. In some embodiments, ambient light can be passed through a portion less than all of an AR environment presented within a user's field of view (e.g., a portion of the AR environment co-located with a physical object in the user's real-world environment that is within a designated boundary (e.g., a guardian boundary) configured to be used by the user while they are interacting with the AR environment). For example, a visual user interface element (e.g., a notification user interface element) can be presented at the head-wearable device, and an amount of ambient light (e.g., 15-50% of the ambient light) can be passed through the user interface element such that the user can distinguish at least a portion of the physical

environment over which the user interface element is being displayed.

[0192] In some embodiments, the systems described herein may also include an eye-tracking subsystem designed to identify and track various characteristics of a user's eye(s), such as the user's gaze direction. The phrase “eye tracking” may, in some examples, refer to a process by which the position, orientation, and/or motion of an eye is measured, detected, sensed, determined, and/or monitored. The disclosed systems may measure the position, orientation, and/or motion of an eye in a variety of different ways, including through the use of various optical-based eye-tracking techniques, ultrasound-based eye-tracking techniques, etc. An eye-tracking subsystem may be configured in a number of different ways and may include a variety of different eye-tracking hardware components or other computer-vision components. For example, an eye-tracking subsystem may include a variety of different optical sensors, such as two-dimensional (2D) or 3D cameras, time-of-flight depth sensors, single-beam or sweeping laser rangefinders, 3D LiDAR sensors, and/or any other suitable type or form of optical sensor. In this example, a processing subsystem may process data from one or more of these sensors to measure, detect, determine, and/or otherwise monitor the position, orientation, and/or motion of the user's eye(s).

[0193] FIG. **18** is an illustration of an example system **1800** that incorporates an eye-tracking subsystem capable of tracking a user's eye(s). As depicted in FIG. **18**, system **1800** may include a light source **1802**, an optical subsystem **1804**, an eye-tracking subsystem **1806**, and/or a control subsystem **1808**. In some examples, light source **1802** may generate light for an image (e.g., to be presented to an eye **1801** of the viewer). Light source **1802** may represent any of a variety of suitable devices. For example, light source **1802** can include a two-dimensional projector (e.g., a LCOS display), a scanning source (e.g., a scanning laser), or other device (e.g., an LCD, an LED display, an OLED display, an active-matrix OLED display (AMOLED), a transparent OLED display (TOLED), a waveguide, or some other display capable of generating light for presenting an image to the viewer). In some examples, the image may represent a virtual image, which may refer to an optical image formed from the apparent divergence of light rays from a point in space, as opposed to an image formed from the light ray's actual divergence.

[0194] In some embodiments, optical subsystem **1804** may receive the light generated by light source **1802** and generate, based on the received light, converging light **1820** that includes the image. In some examples, optical subsystem **1804** may include any number of lenses (e.g., Fresnel lenses, convex lenses, concave lenses), apertures, filters, mirrors, prisms, and/or other optical components, possibly in combination with actuators and/or other devices. In particular, the actuators and/or other devices may translate and/or rotate one or more of the optical components to alter one or more aspects of converging light **1820**. Further, various mechanical couplings may serve to maintain the relative spacing and/or the orientation of the optical components in any suitable combination.

[0195] In one embodiment, eye-tracking subsystem **1806** may generate tracking information indicating a gaze angle of an eye **1801** of the viewer. In this embodiment, control subsystem **1808** may control aspects of optical subsystem **1804** (e.g., the angle of incidence of converging light **1820**) based at least in part on this tracking information. Additionally, in some examples, control subsystem **1808** may store and utilize historical tracking information (e.g., a history of the tracking information over a given duration, such as the previous second or fraction thereof) to anticipate the gaze angle of eye **1801** (e.g., an angle between the visual axis and the anatomical axis of eye **1801**). In some embodiments, eye-tracking subsystem **1806** may detect radiation emanating from some portion of eye **1801** (e.g., the cornea, the iris, the pupil, or the like) to determine the current gaze angle of eye **1801**. In other examples, eye-tracking subsystem **1806** may employ a wavefront sensor to track the current location of the pupil.

[0196] Any number of techniques can be used to track eye **1801**. Some techniques may involve illuminating eye **1801** with infrared light and measuring reflections with at least one optical sensor that is tuned to be sensitive to the infrared light. Information about how the infrared light is

reflected from eye **1801** may be analyzed to determine the position(s), orientation(s), and/or motion(s) of one or more eye feature(s), such as the cornea, pupil, iris, and/or retinal blood vessels. [0197] In some examples, the radiation captured by a sensor of eye-tracking subsystem **1806** may be digitized (i.e., converted to an electronic signal). Further, the sensor may transmit a digital representation of this electronic signal to one or more processors (for example, processors associated with a device including eye-tracking subsystem **1806**). Eye-tracking subsystem **1806** may include any of a variety of sensors in a variety of different configurations. For example, eye-tracking subsystem **1806** may include an infrared detector that reacts to infrared radiation. The infrared detector may be a thermal detector, a photonic detector, and/or any other suitable type of detector. Thermal detectors may include detectors that react to thermal effects of the incident infrared radiation.

[0198] In some examples, one or more processors may process the digital representation generated by the sensor(s) of eye-tracking subsystem **1806** to track the movement of eye **1801**. In another example, these processors may track the movements of eye **1801** by executing algorithms represented by computer-executable instructions stored on non-transitory memory. In some examples, on-chip logic (e.g., an application-specific integrated circuit or ASIC) may be used to perform at least portions of such algorithms. As noted, eye-tracking subsystem **1806** may be programmed to use an output of the sensor(s) to track movement of eye **1801**. In some embodiments, eye-tracking subsystem **1806** may analyze the digital representation generated by the sensors to extract eye rotation information from changes in reflections. In one embodiment, eye-tracking subsystem **1806** may use corneal reflections or glints (also known as Purkinje images) and/or the center of the eye's pupil **1822** as features to track over time.

[0199] In some embodiments, eye-tracking subsystem **1806** may use the center of the eye's pupil **1822** and infrared or near-infrared, non-collimated light to create corneal reflections. In these embodiments, eye-tracking subsystem **1806** may use the vector between the center of the eye's pupil **1822** and the corneal reflections to compute the gaze direction of eye **1801**. In some embodiments, the disclosed systems may perform a calibration procedure for an individual (using, e.g., supervised or unsupervised techniques) before tracking the user's eyes. For example, the calibration procedure may include directing users to look at one or more points displayed on a display while the eye-tracking system records the values that correspond to each gaze position associated with each point.

[0200] In some embodiments, eye-tracking subsystem **1806** may use two types of infrared and/or near-infrared (also known as active light) eye-tracking techniques: bright-pupil and dark-pupil eye tracking, which may be differentiated based on the location of an illumination source with respect to the optical elements used. If the illumination is coaxial with the optical path, then eye **1801** may act as a retroreflector as the light reflects off the retina, thereby creating a bright pupil effect similar to a red-eye effect in photography. If the illumination source is offset from the optical path, then the eye's pupil **1822** may appear dark because the retroreflection from the retina is directed away from the sensor. In some embodiments, bright-pupil tracking may create greater iris/pupil contrast, allowing more robust eye tracking with iris pigmentation, and may feature reduced interference (e.g., interference caused by eyelashes and other obscuring features). Bright-pupil tracking may also allow tracking in lighting conditions ranging from total darkness to a very bright environment.

[0201] In some embodiments, control subsystem **1808** may control light source **1802** and/or optical subsystem **1804** to reduce optical aberrations (e.g., chromatic aberrations and/or monochromatic aberrations) of the image that may be caused by or influenced by eye **1801**. In some examples, as mentioned above, control subsystem **1808** may use the tracking information from eye-tracking subsystem **1806** to perform such control. For example, in controlling light source **1802**, control subsystem **1808** may alter the light generated by light source **1802** (e.g., by way of image rendering) to modify (e.g., pre-distort) the image so that the aberration of the image caused by eye **1801** is reduced.

[0202] The disclosed systems may track both the position and relative size of the pupil (since, e.g., the pupil dilates and/or contracts). In some examples, the eye-tracking devices and components (e.g., sensors and/or sources) used for detecting and/or tracking the pupil may be different (or calibrated differently) for different types of eyes. For example, the frequency range of the sensors may be different (or separately calibrated) for eyes of different colors and/or different pupil types, sizes, and/or the like. As such, the various eye-tracking components (e.g., infrared sources and/or sensors) described herein may need to be calibrated for each individual user and/or eye.

[0203] The disclosed systems may track both eyes with and without ophthalmic correction, such as that provided by contact lenses worn by the user. In some embodiments, ophthalmic correction elements (e.g., adjustable lenses) may be directly incorporated into the artificial reality systems described herein. In some examples, the color of the user's eye may necessitate modification of a corresponding eye-tracking algorithm. For example, eye-tracking algorithms may need to be modified based at least in part on the differing color contrast between a brown eye and, for example, a blue eye.

[0204] FIG. 19 is a more detailed illustration of various aspects of the eye-tracking subsystem illustrated in FIG. 18. As shown in this figure, an eye-tracking subsystem 1900 may include at least one source 1904 and at least one sensor 1906. Source 1904 generally represents any type or form of element capable of emitting radiation. In one example, source 1904 may generate visible, infrared, and/or near-infrared radiation. In some examples, source 1904 may radiate non-collimated infrared and/or near-infrared portions of the electromagnetic spectrum towards an eye 1902 of a user. Source 1904 may utilize a variety of sampling rates and speeds. For example, the disclosed systems may use sources with higher sampling rates in order to capture fixational eye movements of a user's eye 1902 and/or to correctly measure saccade dynamics of the user's eye 1902. As noted above, any type or form of eye-tracking technique may be used to track the user's eye 1902, including optical-based eye-tracking techniques, ultrasound-based eye-tracking techniques, etc.

[0205] Sensor 1906 generally represents any type or form of element capable of detecting radiation, such as radiation reflected off the user's eye 1902. Examples of sensor 1906 include, without limitation, a charge coupled device (CCD), a photodiode array, a complementary metal-oxide-semiconductor (CMOS) based sensor device, and/or the like. In one example, sensor 1906 may represent a sensor having predetermined parameters, including, but not limited to, a dynamic resolution range, linearity, and/or other characteristic selected and/or designed specifically for eye tracking.

[0206] As detailed above, eye-tracking subsystem 1900 may generate one or more glints. As detailed above, a glint 1903 may represent reflections of radiation (e.g., infrared radiation from an infrared source, such as source 1904) from the structure of the user's eye. In various embodiments, glint 1903 and/or the user's pupil may be tracked using an eye-tracking algorithm executed by a processor (either within or external to an artificial reality device). For example, an artificial reality device may include a processor and/or a memory device in order to perform eye tracking locally and/or a transceiver to send and receive the data necessary to perform eye tracking on an external device (e.g., a mobile phone, cloud server, or other computing device).

[0207] FIG. 19 shows an example image 1905 captured by an eye-tracking subsystem, such as eye-tracking subsystem 1900. In this example, image 1905 may include both the user's pupil 1908 and a glint 1910 near the same. In some examples, pupil 1908 and/or glint 1910 may be identified using an artificial-intelligence-based algorithm, such as a computer-vision-based algorithm. In one embodiment, image 1905 may represent a single frame in a series of frames that may be analyzed continuously in order to track the eye 1902 of the user. Further, pupil 1908 and/or glint 1910 may be tracked over a period of time to determine a user's gaze.

[0208] In one example, eye-tracking subsystem 1900 may be configured to identify and measure the inter-pupillary distance (IPD) of a user. In some embodiments, eye-tracking subsystem 1900 may measure and/or calculate the IPD of the user while the user is wearing the artificial reality

system. In these embodiments, eye-tracking subsystem 1900 may detect the positions of a user's eyes and may use this information to calculate the user's IPD.

[0209] As noted, the eye-tracking systems or subsystems disclosed herein may track a user's eye position and/or eye movement in a variety of ways. In one example, one or more light sources and/or optical sensors may capture an image of the user's eyes. The eye-tracking subsystem may then use the captured information to determine the user's inter-pupillary distance, interocular distance, and/or a 3D position of each eye (e.g., for distortion adjustment purposes), including a magnitude of torsion and rotation (i.e., roll, pitch, and yaw) and/or gaze directions for each eye. In one example, infrared light may be emitted by the eye-tracking subsystem and reflected from each eye. The reflected light may be received or detected by an optical sensor and analyzed to extract eye rotation data from changes in the infrared light reflected by each eye.

[0210] The eye-tracking subsystem may use any of a variety of different methods to track the eyes of a user. For example, a light source (e.g., infrared light-emitting diodes) may emit a dot pattern onto each eye of the user. The eye-tracking subsystem may then detect (e.g., via an optical sensor coupled to the artificial reality system) and analyze a reflection of the dot pattern from each eye of the user to identify a location of each pupil of the user. Accordingly, the eye-tracking subsystem may track up to six degrees of freedom of each eye (i.e., 3D position, roll, pitch, and yaw) and at least a subset of the tracked quantities may be combined from two eyes of a user to estimate a gaze point (i.e., a 3D location or position in a virtual scene where the user is looking) and/or an IPD.

[0211] In some cases, the distance between a user's pupil and a display may change as the user's eye moves to look in different directions. The varying distance between a pupil and a display as viewing direction changes may be referred to as “pupil swim” and may contribute to distortion perceived by the user as a result of light focusing in different locations as the distance between the pupil and the display changes. Accordingly, measuring distortion at different eye positions and pupil distances relative to displays and generating distortion corrections for different positions and distances may allow mitigation of distortion caused by pupil swim by tracking the 3D position of a user's eyes and applying a distortion correction corresponding to the 3D position of each of the user's eyes at a given point in time. Thus, knowing the 3D position of each of a user's eyes may allow for the mitigation of distortion caused by changes in the distance between the pupil of the eye and the display by applying a distortion correction for each 3D eye position. Furthermore, as noted above, knowing the position of each of the user's eyes may also enable the eye-tracking subsystem to make automated adjustments for a user's IPD.

[0212] In some embodiments, a display subsystem may include a variety of additional subsystems that may work in conjunction with the eye-tracking subsystems described herein. For example, a display subsystem may include a varifocal subsystem, a scene-rendering module, and/or a vergence-processing module. The varifocal subsystem may cause left and right display elements to vary the focal distance of the display device. In one embodiment, the varifocal subsystem may physically change the distance between a display and the optics through which it is viewed by moving the display, the optics, or both. Additionally, moving or translating two lenses relative to each other may also be used to change the focal distance of the display. Thus, the varifocal subsystem may include actuators or motors that move displays and/or optics to change the distance between them. This varifocal subsystem may be separate from or integrated into the display subsystem. The varifocal subsystem may also be integrated into or separate from its actuation subsystem and/or the eye-tracking subsystems described herein.

[0213] In one example, the display subsystem may include a vergence-processing module configured to determine a vergence depth of a user's gaze based on a gaze point and/or an estimated intersection of the gaze lines determined by the eye-tracking subsystem. Vergence may refer to the simultaneous movement or rotation of both eyes in opposite directions to maintain single binocular vision, which may be naturally and automatically performed by the human eye. Thus, a location where a user's eyes are verged is where the user is looking and is also typically the location where

the user's eyes are focused. For example, the vergence-processing module may triangulate gaze lines to estimate a distance or depth from the user associated with intersection of the gaze lines. The depth associated with intersection of the gaze lines may then be used as an approximation for the accommodation distance, which may identify a distance from the user where the user's eyes are directed. Thus, the vergence distance may allow for the determination of a location where the user's eyes should be focused and a depth from the user's eyes at which the eyes are focused, thereby providing information (such as an object or plane of focus) for rendering adjustments to the virtual scene.

[0214] The vergence-processing module may coordinate with the eye-tracking subsystems described herein to make adjustments to the display subsystem to account for a user's vergence depth. When the user is focused on something at a distance, the user's pupils may be slightly farther apart than when the user is focused on something close. The eye-tracking subsystem may obtain information about the user's vergence or focus depth and may adjust the display subsystem to be closer together when the user's eyes focus or verge on something close and to be farther apart when the user's eyes focus or verge on something at a distance.

[0215] The eye-tracking information generated by the above-described eye-tracking subsystems may also be used, for example, to modify various aspect of how different computer-generated images are presented. For example, a display subsystem may be configured to modify, based on information generated by an eye-tracking subsystem, at least one aspect of how the computer-generated images are presented. For instance, the computer-generated images may be modified based on the user's eye movement, such that if a user is looking up, the computer-generated images may be moved upward on the screen. Similarly, if the user is looking to the side or down, the computer-generated images may be moved to the side or downward on the screen. If the user's eyes are closed, the computer-generated images may be paused or removed from the display and resumed once the user's eyes are back open.

[0216] The above-described eye-tracking subsystems can be incorporated into one or more of the various artificial reality systems described herein in a variety of ways. For example, one or more of the various components of system **1800** and/or eye-tracking subsystem **1900** may be incorporated into any of the augmented-reality systems in and/or virtual-reality systems described herein in to enable these systems to perform various eye-tracking tasks (including one or more of the eye-tracking operations described herein).

[0217] The process parameters and sequence of the steps described and/or illustrated herein are given by way of example only and can be varied as desired. For example, while the steps illustrated and/or described herein may be shown or discussed in a particular order, these steps do not necessarily need to be performed in the order illustrated or discussed. The various exemplary methods described and/or illustrated herein may also omit one or more of the steps described or illustrated herein or include additional steps in addition to those disclosed.

[0218] The preceding description has been provided to enable others skilled in the art to best utilize various aspects of the exemplary embodiments disclosed herein. This exemplary description is not intended to be exhaustive or to be limited to any precise form disclosed. Many modifications and variations are possible without departing from the spirit and scope of the present disclosure. The embodiments disclosed herein should be considered in all respects illustrative and not restrictive. Reference should be made to the appended claims and their equivalents in determining the scope of the present disclosure.

[0219] Unless otherwise noted, the terms “connected to” and “coupled to” (and their derivatives), as used in the specification and claims, are to be construed as permitting both direct and indirect (i.e., via other elements or components) connection. In addition, the terms “a” or “an,” as used in the specification and claims, are to be construed as meaning “at least one of.” Finally, for ease of use, the terms “including” and “having” (and their derivatives), as used in the specification and claims, are interchangeable with and have the same meaning as the word “comprising.”

Claims

1. A device comprising: a plurality of waveguides consisting of a first plurality of waveguides, a second plurality of waveguides and a third plurality of waveguides; and an optical dispatching circuit comprising a first input, a second input and a third input, the optical dispatching circuit configured to direct light from the first input through each of the first plurality of waveguides, to direct light from the second input through each of the second plurality of waveguides, and to direct light from the third input through each of the third plurality of waveguides.
2. The device of claim 1, wherein the optical dispatching circuit comprises: an input multiplexer coupled to the first input, the second input and the third input and comprising an output; a bus waveguide coupled to the output of the input multiplexer; and a transport slab arranged adjacent to and apart from the bus waveguide, the transport slab coupled to each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.
3. The device of claim 2, wherein a distance between the bus waveguide and the transport slab narrows from a first end of the bus waveguide proximate to the input multiplexer to an opposing end of the bus waveguide distal from the input multiplexer.
4. The device of claim 2, further comprising a plurality of output multiplexers that couple the transport slab to the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides, wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.
5. The device of claim 2, wherein the input multiplexer comprises: a first multiplexer waveguide coupled to the first input; a second multiplexer waveguide coupled to the second input; and a third multiplexer waveguide coupled to the third input and coupled to the output of the input multiplexer, wherein the first multiplexer waveguide widens from the first input to the output of the input multiplexer, wherein the second multiplexer waveguide narrows from the second input to the output of the input multiplexer, and wherein the third multiplexer waveguide narrows from the third input to the output of the input multiplexer.
6. The device of claim 1, wherein the optical dispatching circuit comprises a waveguide tree coupled to the first input, the second input and the third input, and comprising a plurality of outputs that includes an output for each of the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides.
7. The device of claim 6, wherein the waveguide tree comprises a plurality of Y-shaped waveguide sections.
8. The device of claim 6, wherein the waveguide tree comprises a plurality of directional couplers, each directional coupler configured to swap spatial modes of light across a pair of waveguides therein.
9. The device of claim 8, wherein, in each directional coupler of the plurality of directional couplers, one of the pair of waveguides tapers narrower across the directional coupler from an input side to an output side while another of the pair of waveguides tapers wider from the input side to the output side of the directional coupler.
10. The device of claim 1, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.
11. The device of claim 10, further comprising a plurality of output multiplexers that couple each of the plurality of receiving waveguides to the first plurality of waveguides, the second plurality of waveguides and the third plurality of waveguides, wherein each receiving waveguide of the plurality of receiving waveguides is coupled to a respective output multiplexer of the plurality of

output multiplexers, and wherein each output multiplexer of the plurality of output multiplexers is coupled to one of the first plurality of waveguides, one of the second plurality of waveguides, and one of the third plurality of waveguides.

12. The device of claim 10, wherein a width of each receiving waveguide of the plurality of receiving waveguides increases with distance from a central axis of the coupler waveguide.

13. The device of claim 10, further comprising: a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide; a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide; a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide, wherein the first tip, the second tip and the third tip each have a width between 0.9 μm and 1.1 μm .

14. The device of claim 1, further comprising: a first light source arranged to direct light of a first wavelength through the first input of the optical dispatching circuit; a second light source arranged to direct light of a second wavelength, different to the first wavelength, through the second input of the optical dispatching circuit; and a third light source arranged to direct light of a third wavelength, different to the first wavelength and the second wavelength, through the third input of the optical dispatching circuit.

15. The device of claim 14, wherein the first wavelength is between 600 nm and 700 nm, the second wavelength is between 500 nm and 600 nm, and the third wavelength is between 400 nm and 500 nm.

16. The device of claim 1, wherein each of the plurality of waveguides comprises a plurality of emitter structures configured to output light.

17. A display comprising the device of claim 16 and a liquid crystal on silicon (LCoS) panel arranged to receive light output from the plurality of emitter structures.

18. A method comprising: directing light from a first light source into a first input of an optical dispatching circuit, through the optical dispatching circuit, into a first plurality of waveguides and into a first plurality of emitter structures arranged within the first plurality of waveguides; directing light from a second light source into a second input of the optical dispatching circuit, through the optical dispatching circuit, into a second plurality of waveguides and into a second plurality of emitter structures arranged within the second plurality of waveguides; and directing light from a third light source into a third input of the optical dispatching circuit, through the optical dispatching circuit, into a third plurality of waveguides and into a third plurality of emitter structures arranged within the third plurality of waveguides.

19. The method of claim 18, wherein the optical dispatching circuit comprises a coupler waveguide arranged to receive light from each of the first input, the second input and the third input in an input region and to output the light from each of the first input, second input and the third input into a plurality of receiving waveguides arranged radially from the input region.

20. The method of claim 19, wherein: directing light from the first light source into the first input of the optical dispatching circuit comprises directing the light from the first light source through a first input waveguide coupled to the first input and terminating with a first tip proximate to the coupler waveguide; directing light from the second light source into the second input of the optical dispatching circuit comprises directing the light from the second light source through a second input waveguide coupled to the second input and terminating with a second tip proximate to the coupler waveguide; and directing light from the third light source into the third input of the optical dispatching circuit comprises directing the light from the third light source through a third input waveguide coupled to the third input and terminating with a third tip proximate to the coupler waveguide.
